A Dual-Track Adaptive Projection Algorithm for a Nonlinear Split Feasibility Problem with Mixed Convex–Pseudoconvex Constraints

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Revised manuscript with changes marked. This version compares the revised manuscript with the originally submitted one. Added text is highlighted in yellow; ~~deleted text is struck through in red~~. New tables and figures are shown on a yellow background, and deleted or replaced tables and figures are framed in red and labelled "Former Table/Figure" with their original numbers. Paragraphs that were moved are shown once, at their new place, compared with their original wording. Section, equation, theorem and table numbers are those of the revised manuscript; numbers in deleted text refer to the submitted version. Because the reference list below also contains the deleted references, citation numbers differ from those of the clean revised manuscript. A bar in the left margin marks every changed passage.

~~**ABSTRACT**~~

~~We study a nonlinear split feasibility problem whose output set is a normal box and whose component maps split into convex and pseudoconvex blocks. Classical *CQ* solvers presuppose a linear operator, while existing self-adaptive methods require the aggregate proximity function to be convex, which a single pseudoconvex component invalidates. A Tchebyshev scalarization turns feasibility into a closed-form max-type violation function. We minimize it with the Dual-Track Adaptive Projection (DTAP) algorithm, which takes a residual-adaptive step on the convex constraints and a safeguarded diminishing step on the pseudoconvex ones, and projects each iterate onto the base set. Its step sizes need no operator-norm, Lipschitz, or Hölder modulus, yet we prove strong convergence to a feasible point. As an application, we recast two disjoint feature-selection models, one fractional-programming and one fuzzy-rough, as one nonlinear SFP, with no artificial terms; a sparse variant DTAP-S adds a feature budget. On eight UCI benchmarks DTAP matches or improves accuracy and AUC over the base models while reaching feasibility in fewer iterations, and DTAP-S yields compact, stable selections.~~

ABSTRACT

We study a nonlinear split feasibility problem with a normal-box output set and differentiable convex and pseudoconvex component functions. A max-type violation measure identifies nearly active constraints in two blocks. The Dual-Track Adaptive Projection (DTAP) algorithm combines a residual-scaled step for the convex block with a safeguarded diminishing step for the pseudoconvex block, followed by

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projection onto the base convex set. Its step sizes require no prescribed operator norm, Lipschitz constant, or Hölder modulus. Under consistency and continuous differentiability, we prove convergence of the whole sequence to a feasible point in finite-dimensional Euclidean space. The proof uses local bounds on the bounded iterate region. The proportional mixing weight is a residual-responsive choice, without a claim of optimality. A feature-selection formulation combines criteria from fractional and fuzzy-rough models and belongs to the convex subclass. Experiments report constraint activity, classification comparisons, and subgradient comparisons with empirically tuned moduli. Controlled examples with a nonconvex pseudoconvex constraint test the pseudoconvex track. On these examples, DTAP is sensitive to the scale of its diminishing step. The results also show initialization sensitivity and do not establish uniform computational superiority.

KEYWORDS
Split feasibility problem; Pseudoconvex optimization; Tchebyshev scalarization; Adaptive ~~(Lipschitz-free)~~ step size; Strong convergence; Feature selection

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1. Introduction

~~The *split feasibility problem* (SFP), introduced by Censor and Elfving [6], seeks a point $x^\star \in C \subseteq \mathbb{R}^n$ whose image under a bounded linear operator $\mathcal{A} : \mathbb{R}^n \to \mathbb{R}^m$ lies in a convex set $Q \subseteq \mathbb{R}^m$:~~

$$\text{~~find } x^\star \in C \text{ such that } \mathcal{A}x^\star \in Q.\text{~~}$$

~~It is a general formulation that specializes to convex feasibility, linear inverse problems, image reconstruction [5], signal processing [9], and radiotherapy planning [7]. Its canonical solver is Byrne's CQ method [5],~~

$$x^{k+1} = P_C\Big(x^k - \tau\,\mathcal{A}^\top(I - P_Q)\mathcal{A}x^k\Big), \qquad \tau \in \big(0,\, 2/\|\mathcal{A}\|^2\big).$$

~~This solver has two drawbacks, namely a step size depending on the operator norm $\|\mathcal{A}\|$ and a projection P_Q with no closed form for generic Q. The second is met by relaxed-projection variants [15, 36], which replace Q by an outer half-space, and the first by the self-adaptive rule of López et al. [25], which calibrates the step from the current value of the merit function $f(x) := \frac{1}{2}\|(I - P_Q)\mathcal{A}x\|^2$,~~

$$\tau_n = \frac{\rho_n\, f(x_n)}{\|\nabla f(x_n)\|^2}, \qquad 0 < \rho_n < 4.$$

~~Rule (3) is the reference point of this paper, so we isolate the mechanism behind it, beyond its removal of $\|\mathcal{A}\|$. Its analysis turns on the single inequality $\langle \nabla f(x),\, x - z\rangle \geq 2f(x)$, valid at every solution z because $I - P_Q$ is firmly nonexpansive; this inequality delivers both the admissible range $\rho_n \in (0, 4)$ and the per-step decrease $\|x_{n+1} - z\|^2 \leq$~~

~~$\|x_n - z\|^2 - \rho_n(4-\rho_n)f^2(x_n)/\|\nabla f(x_n)\|^2$. Strong convergence is available in Hilbert space through regularized and Krasnosel'skiĭ–Mann variants [34]. The mechanism above is therefore a property of the *composition* $(I-P_Q)\mathcal{A}$: it comes from the linearity of $\mathcal{A}$, and the residual calibration alone does not provide it.~~

The split feasibility problem (SFP) models constrained reconstruction [6]. For closed convex sets $C \subseteq \mathbb{R}^n$, $Q \subseteq \mathbb{R}^m$, and a matrix $\mathcal{A}$, it seeks

$$\text{find } x^\star \in C \quad \text{such that} \quad \mathcal{A}x^\star \in Q. \tag{1}$$

Byrne's CQ algorithm [5] uses

$$x^{k+1} = P_C\big(x^k - \tau\mathcal{A}^\top(I-P_Q)\mathcal{A}x^k\big), \qquad 0 < \tau < 2/\|\mathcal{A}\|^2. \tag{2}$$

Supporting half-spaces simplify projections [36], while self-adaptive steps reduce parameter dependence [25, 38, 40]. Operator splitting [39] and modified alternating directions [35] provide decomposition strategies. Adaptive steps and decomposition are therefore established principles.

We study the nonlinear SFP

$$\text{find } x^\star \in K \quad \text{such that} \quad F(x^\star) \in Q, \tag{3}$$

where $K \subseteq \mathbb{R}^n$ is closed and convex, $Q = \prod_{i=1}^m (-\infty, b_i]$, and each differentiable component of $F = (F_1, \dots, F_m)$ is convex or pseudoconvex on K. Pseudoconvex functions have convex sublevel sets, so the feasible set is convex even when some components are nonconvex. We seek feasibility, rather than minimization of every component.

Nonlinear SFP and generalized convex feasibility have several relevant antecedents. Li et al. [23] use a convex aggregate proximity function, a property not ensured by componentwise pseudoconvexity. Yen et al. [37] combine equilibrium and nonlinear image constraints. Hu et al. [19] treat quasiconvex inequalities, which include our component class. Our method exploits the additional derivative information of convex components within this established feasibility setting.

The box gives the explicit violation measure

$$J(x) = d_\infty(F(x), Q) = \max_{1\le i\le m} (F_i(x) - b_i)^+.$$

The Dual-Track Adaptive Projection algorithm (DTAP) forms violation-weighted gradient sums from nearly maximal violations in each block. Convex gradient inequalities justify a residual-scaled Polyak-type step [25, 29], whereas pseudoconvexity supports a safeguarded diminishing step. A weighted combination is projected onto K. The proportional weight responds to block violations and is not claimed to be unique or optimal. Its role in preventing neglect is used in the convergence proof. A one-step bound valid for all weights in $[0, 1]$ does not by itself establish convergence for every weight sequence. No prescribed operator norm, Lipschitz constant, or Hölder modulus is required, although the proof uses continuity and local bounds on the bounded iterate region. The analysis uses a quasi-Fejér estimate to establish boundedness before invoking these local bounds. Continuous gradients need not be Lipschitz.

Feature-selection criteria from fractional and fuzzy-rough models [32, 33] define separate threshold constraints. This construction expresses simultaneous acceptance, not equivalence to the source optimization models. The positive-semidefinite quadratic-over-

positive-affine criterion is convex, so this application belongs to the convex subclass. Controlled examples with a nonconvex pseudoconvex constraint test the pseudoconvex track. Classification scores feature selections that are computed once and then reused, on the same holdout split that was fixed when they were computed, and tuned subgradient methods can match or outperform DTAP.

The contributions are as follows.

C1. A dual-track method combines convex decrease estimates and pseudoconvex sign information through an adaptive weight.

C2. Under consistency, smoothness, and the stated step conditions, the whole sequence converges to a feasible point in finite-dimensional Euclidean space. Convergence of distances and one feasible cluster point suffice.

C3. Experiments distinguish prediction from feasibility cost and report constraint activity, tuned subgradient comparisons, and initialization and scaling sensitivity. They do not establish uniform superiority.

~~Linearity fails in many contemporary problems, among them the dose maps of intensity-modulated radiotherapy, where the differing densities of organs and tissues make the beam-to-dose relation nonlinear [23], economic equilibrium models [37], and the training of neural predictors [31]. This leads to the *nonlinear split feasibility problem* (NSFP): given closed convex $K \subseteq \mathbb{R}^n$ and $Q \subseteq \mathbb{R}^m$ and a nonlinear map $F : \mathbb{R}^n \to \mathbb{R}^m$,~~

$$\text{~~find } x^\star \in K \text{ ~~such that~~ } F(x^\star) \in Q.$$

~~The NSFP is not itself new. Li et al. [23] treat its multiple-sets form by minimizing a smooth Euclidean proximity function with a self-adaptive, backtracked step, while Yen et al. [37] minimize the max-type merit $\max_i |(I - P_{Q_i})F_i(x)|^2$ over the solution set of an equilibrium problem, driving it by the Clarke subdifferential on the active index set, which is the same scalarization and active-set device we use below. Both papers replace, by a structural hypothesis, the quantitative inequality that linearity used to supply for free: [23] require the aggregate proximity function to be convex, so that its gradient is co-coercive, and [37] require every component map to be *quasilinear*, that is, both quasiconvex and quasiconcave. Neither hypothesis is ours, and neither of these classes coincides with the one we assume. Global convexity of the proximity function fails the moment a single component is only pseudoconvex; two-sided quasilinearity already excludes an ordinary convex map such as $x \mapsto x^2$, which is quasiconvex but not quasiconcave. We ask instead that each component be convex or pseudoconvex, a one-sided condition that these two classes neither contain nor are contained in, as Section 2.2 makes precise. Two further difficulties attend the nonlinear case. When some F_i is nonconvex the global gradient inequality is lost, so CQ-type separating hyperplanes may cut off the solution; and the available step sizes are tied either to a Lipschitz constant of ∇F or, for the wider quasi-convex class, to a Hölder modulus [19], both hard to estimate at scale. Section 3.4 makes these challenges precise as (D1)–(D4). A complementary motivation comes from *feature selection* [16, 30], where embedded models cast selection as a constrained continuous program over a feature-weight vector. Two validated but isolated models live here: the fractional-programming model of Wang et al. [32], which minimizes a redundancy-to-relevance ratio and is solved by a projection neural network, and the fuzzy-rough model AFRSFCV [33], which fuses class variance into fuzzy lower approximations and is solved by a softmax parametrization. Optimizing either in isolation tends to yield overly sparse selections, and neither offers a common~~

language for combining criteria.

These two lines of work, nonlinear split feasibility on the one hand and embedded feature selection on the other, turn out to be two instances of a single problem. This paper addresses three questions.

(Q1) Is there a structured but broadly applicable subclass of NSFP for which one can design an iterative method that is simultaneously (i) free of operator-norm, Lipschitz, and Hölder constants; (ii) free of projections onto the intractable solution set; and (iii) provably *strongly* convergent?

(Q2) Can heterogeneous selection criteria, some convex and some only pseudoconvex, be expressed in one feasibility model *without inventing new penalty terms*, so that the model inherits the empirical meaning of its constituents?

(Q3) Does solving feature selection *as* a feasibility problem offer any concrete advantage over solving each constituent model on its own, whether in selection quality, stability, or computational cost?

We answer all three questions affirmatively. On the theoretical side we isolate the subclass of NSFP in which the output set is a *normal box* $Q = (-\infty, b_1] \times \cdots \times (-\infty, b_m]$ and the component maps split into a convex group I^{cvx} and a pseudoconvex group I^{ps}. The box structure admits a *Tchebyshev scalarization*: measuring the residual in the sup-norm collapses $d_\infty(F(x), Q)$ into the closed-form max-type function $J(x) = \max_i (F_i(x) - b_i)^+$, which certifies feasibility one constraint at a time and whose maximizers expose the binding constraints. The split into convex and pseudoconvex groups then leads to a *dual-track* step-size policy that reads exactly this information, described next.

We work in finite-dimensional Euclidean space with smooth component maps and a nonempty feasible set; these mild conditions are made precise in Assumption 1. Our target is *feasibility*, that is, returning a point of Ω, rather than the minimization of a designated objective. Any optimization criterion of interest enters the model as one of the constraint functions F_i, so that 'solving' the problem means bringing all criteria below their thresholds. This feasibility viewpoint allows several heterogeneous criteria, convex and pseudoconvex alike, to coexist in a single problem, and it lets a single algorithm with a single parameter setting serve the whole study. The price is that we do not claim optimality of the returned point for any one criterion; we return a point that is acceptable for all of them, which is the requirement in the feature-selection application.

This paper makes five principal contributions.

C1. A tractable nonlinear split-feasibility model. We formulate the *normal-box* NSFP with mixed convex–pseudoconvex components and derive, from a Tchebyshev scalarization, a closed-form max-type violation function J whose zero set coincides with the feasible set. This turns an abstract feasibility requirement into a single nonsmooth merit function whose maximizers identify the binding constraints, which is the structure the dual-track step exploits (Section 3).

C2. The DTAP algorithm with Lipschitz-free adaptive steps. We propose the Dual-Track Adaptive Projection algorithm, which decomposes the constraint index set into a convex track and a pseudoconvex track and assigns each the step rule that fits its geometry. On the convex track we extend rule (3) to the nonlinear setting: the step is $\lambda \Phi_k^{\mathrm{cvx}} / \left\| g_k^{\mathrm{cvx}} \right\|^2$ with $\lambda = 2\rho_{\mathrm{cvx}} \in (0, 4)$, where Φ^{cvx} is a partial quadratic violation and g^{cvx} its gradient. Equivalently it is a Polyak step [29] for Φ^{cvx} with known optimal value zero, but with the *doubled* relaxation range of López et al. rather than the classical $(0, 2)$; the doubling is no longer licensed by

firm nonexpansiveness, which linearity would have supplied; it now rests on the aggregated inequality $\langle g_k^{\mathrm{cvx}}, x^k - x^\star \rangle \geq 2\Phi_k^{\mathrm{cvx}}$ that convexity of the individual F_i still yields. On the pseudoconvex track, where only sign information remains, we use a *safeguarded diminishing step*. The two directions are merged by a soft proportional weight, and each iteration ends with a single projection onto the base set K. No operator norm, Lipschitz, or Hölder constant is needed, and the method projects only onto the base set, never onto the output set Q or the intractable solution set (Section 3).

C3. **A strong-convergence theorem.** We prove that the DTAP iterates converge *strongly* to a feasible point under three mild assumptions (Theorem 3.12). The proof rests on a novel one-step quasi-Fejér inequality that yields boundedness, distance convergence, and decay of the convex-track violation; feasibility on the pseudoconvex track is then secured by a contradiction argument driven by continuity and pseudoconvexity. The purely convex and purely pseudoconvex regimes follow as corollaries, and a best-violation rate of $O(N^{-1/3})$ is obtained (Section 3).

C4. **A unified, constraint-faithful feature-selection framework.** We show that the fractional-programming model [32] and the fuzzy-rough-set model AFRS-FCV [33] reduce to a *single* normal-box NSFP whose three constraint functions are taken *unchanged* from the two models' own objectives, namely two convex terms from the fuzzy-rough model and one pseudoconvex ratio from the fractional model. No artificial penalty is introduced, so the unified model preserves the empirical semantics of its constituents while becoming directly solvable by DTAP; new criteria can be appended simply as further convex or pseudoconvex constraints, and a sparse-projection variant DTAP-S adds an explicit, tunable feature budget (Section 4.1).

C5. **Empirical validation.** On eight UCI benchmarks spanning a wide range of sample sizes, dimensions, class counts, and balance levels, and against four baselines (the two base models and two subgradient variants for quasi-convex feasibility [19]), DTAP preserves or improves accuracy and AUC over the base models, with the gains concentrated on the SVM classifier and significant at the nominal level under Wilcoxon and Friedman tests, and it produces stable, consistent feature sets. Against the same-class subgradient methods DTAP yields nearly identical selections yet reaches feasibility in substantially fewer iterations and lower wall-clock time, up to three orders of magnitude on the larger datasets, while avoiding the Hölder-modulus estimation those methods require (Section 4).

The paper is organized as follows. Section 2 reviews the literature and positions DTAP against twelve representative prior works. Section 2 collects the analytical preliminaries. Section 3 states the normal-box NSFP, derives the Tchebyshev merit function, presents the dual-track decomposition and the standing assumptions, specifies DTAP, and then develops the convergence analysis. Section 4.1 builds the unified feature-selection framework. Section 4 reports the numerical study, and Section 5 concludes with limitations and extensions.

Section 2 gives preliminaries. Section 3 presents DTAP and its convergence analysis. Section 4 reports experiments, and Section 5 concludes.

[**Former Section 2**] **Related work**

We review four threads (linear split feasibility, nonlinear and quasi-convex feasibility, optimization-based feature selection, and decomposition) and position DTAP in Table 1.

[**Former Section 2.1**] **Linear split feasibility**

Introduced by Censor and Elfving [6], the SFP was simplified by Byrne's *CQ* method [5], which uses only P_C, P_Q and a gradient step through $\mathcal{A}^\top(I-P_Q)\mathcal{A}$ but pays with a step size tied to $\|\mathcal{A}\|^2$ and a projection P_Q explicit only for structured Q. Two lines address these costs: relaxed projection methods [15, 36] replace Q by an outer half-space with closed-form projection, and the self-adaptive rule (3) of López et al. [25] calibrates the step from the current residual, removing $\|\mathcal{A}\|$. Xu [34] proved *strong* convergence for regularized and Krasnosel'skiĭ–Mann variants in Hilbert space.

Rule (3) has since been extended beyond convex sets. Chen et al. [8] keep it verbatim, as $\rho_k = \lambda_k f_k(u_k)/\|\nabla f_k(u_k)\|^2$ over relaxed level sets $C_{k,\xi}, Q_{k,\xi}$, and allow the level-set functions c, q to be continuous but nonconvex by replacing the subdifferential with an *S-subdifferential*: a modulus $r_f > 0$ must exist for which $f + \frac{r_f}{2} d_S^2$ is subdifferentiable, equivalently $\partial_{S,r_f} f(u) = \partial f(u) + r_f(I - P_S)(u)$. This is a genuine relaxation of convexity, but it is neither ours nor comparable to ours: a pseudoconvex F_j need not admit such an r_f, and a function that does need not be pseudoconvex. What matters more for the present paper is where the nonconvexity is placed. In [8] the operator remains a *matrix*, so the merit function $f_k(u) = \frac{1}{2}\left\|\mathcal{A}u - P_{Q_{k,\xi}}\mathcal{A}u\right\|^2$ is a convex composition and the subgradient inequality that drives their proof is available for free; the nonconvexity sits in the sets, never in the map. Their relaxation range is accordingly $\lambda_k \in (0, 2)$, obtained from the plain gradient inequality rather than from the co-coercivity that yields López et al.'s $(0, 4)$. The residual-based, constant-free philosophy of this thread and its strong-convergence target inform our design; the mechanism that licenses it does not carry over to a nonlinear F, and Section 2.2 takes up what replaces it.

[**Former Section 2.2**] **Nonlinear and quasi-convex feasibility**

When the map is nonlinear, the global gradient inequality fails and *CQ*-type hyperplanes may exclude the solution. The closest predecessor is Li et al. [23], who pose the nonlinear *multiple-sets* SFP (find $x \in \bigcap_i C_i$ with $F(x) \in \bigcap_j Q_j$ for a continuously differentiable F) and minimize over a bounded convex $\Omega \supseteq \bigcap_i C_i$ the Euclidean proximity function $p(x) = \frac{1}{2}\sum_i \alpha_i \|(I - P_{C_i})x\|^2 + \frac{1}{2}\sum_j \beta_j \left\|(I - P_{Q_j})F(x)\right\|^2$ by the projected step $x^{k+1} = P_\Omega[x^k - \beta_{k+1}\nabla p(x^k)]$, the step β_{k+1} being chosen self-adaptively by backtracking. That framework does not sit beside our geometry; it contains it. A normal box is the intersection $\bigcap_{i=1}^m \{z \in \mathbb{R}^m : z_i \le b_i\}$ of m half-spaces, so with $C_1 = K$ and the componentwise violation $v_i(x) := (F_i(x) - b_i)^+$ one has $\|(I - P_{Q_i})F(x)\| = v_i(x)$ and therefore

$$p(x) = \frac{1}{2}\sum_{i=1}^m \beta_i\, v_i(x)^2, \qquad \nabla p(x) = \sum_{i=1}^m \beta_i\, v_i(x)\, \nabla F_i(x).$$

Comparing (5) with the DTAP direction (14) below, g^{cvx} is exactly ∇p with unit weights, restricted to the δ-active convex indices. The direction is therefore not where the novelty lies, and we do not claim it there; the novelty lies in what makes the step legitimate. The convergence proof of [23] runs entirely on co-coercivity of ∇p, which their Lemma 2.4 obtains from *convexity of p* together with a Lipschitz ∇p on the bounded Ω; co-coercivity also keeps the right-hand side of their backtracking test nonnegative, and hence makes the search terminate at all. On a normal box, p is convex precisely when every F_i is convex, since then $v_i = (F_i - b_i)^+$ is convex and nonnegative and $t \mapsto t^2$ is convex and increasing on $[0, \infty)$. A single pseudoconvex component invalidates these implications: v_j^2 need not be convex, p need not be convex, and ∇p

need not be monotone, so the right-hand side of their backtracking test may become negative. [Former footnote: In the notation of Eq. (3.3) of [23], the acceptance test requires $\beta_{k+1}\left\|\nabla p(x^k)-\nabla p(x^{k+1})\right\|^2 \le (2-\mu)\left\langle x^k-x^{k+1}, \nabla p(x^k)-\nabla p(x^{k+1})\right\rangle$; its right-hand side is nonnegative for every pair of iterates iff ∇p is monotone, which on a normal box holds iff every F_i is convex.] When it does, no positive step size satisfies the test, and the line search is not well-defined. DTAP replaces the global and unverifiable hypothesis 'p is convex' by a classification that can be checked one constraint at a time, and replaces the search by a closed-form step; it also drops the requirements that Ω be bounded and that $F, \nabla F$ be globally Lipschitz, deriving boundedness of the iterates instead (Lemma 3.9).

The other close nonlinear predecessor is Yen et al. [37], and the overlap is at the level of the merit function and its differentiation, not merely the topic. On a normal box their componentwise term $p_i(x) = |(I-P_{Q_i})F_i(x)|^2$ equals $v_i(x)^2$ and their gradient $\nabla p_i(x) = 2(I-P_{Q_i})F_i(x)\nabla F_i(x)$ equals $2v_i(x)\nabla F_i(x)$, so their scalarized objective $\max_i p_i$ is the square of our J and their search direction, a convex combination of active ∇p_i selected through the active set $I(x)=\{i : p_i(x)=p(x)\}$ and the Clarke identity $\partial^0 p(x) = \mathrm{co}\{\nabla p_i(x) : i \in I(x)\}$, is exactly the mechanism our tracks (14)–(15) implement. We do not reintroduce that mechanism as novel. Three differences separate the methods, and each is verifiable. First, the admissible map class. Yen et al. require every F_i to be *quasilinear*; we require convex or pseudoconvex. Neither class contains the other: $x \mapsto x^2$ is convex, hence covered here, but not quasiconcave, hence excluded by their (A2); conversely a monotone cubic $x \mapsto x^3$ is quasilinear, hence covered by them, but has a stationary point that is not a minimizer and so is neither convex nor pseudoconvex. Second, the driving step. Their feasibility direction is a normalized subgradient with a diminishing step $\alpha_k = \beta_k/\max\{\delta_k, \|g_k\|\}$ under $\sum \beta_k^2 < \infty$; on our convex track we instead use the residual-adaptive rule (3), whose length is set by the current violation rather than by a prescribed sequence, and which is not diminishing. Third, the source set. Their C is the solution set of a paramonotone equilibrium problem, handled by a second projected-subgradient track, whereas ours is the explicit base set K; this is a deliberate modelling choice, since in our target application K is a simplex or box onto which projection is closed-form, and it lets the second track address the *pseudoconvex output constraints* rather than an implicit source constraint. In short, Yen et al. decompose source-versus-image; DTAP decomposes the image constraints by their individual geometry.

The remaining nonlinear thread is quasi-convex feasibility. Tuan et al. [31] embed split-feasibility constraints in controllable Pareto-front learning, and Hu et al. [19] solve systems of quasi-convex inequalities by subgradient methods with constant and dynamic steps. Their constraint class is strictly *larger* than ours: convex and pseudoconvex functions are quasi-convex, so (6) is one of their instances, and we accordingly use their two variants as baselines in Section 4. This is the opposite relation to the one with Yen et al.: Hu et al. sit outside and above our class, Yen et al. beside it. Generality has a price. A uniform treatment of the whole quasi-convex class leaves only sign information on every constraint, so their dynamic step must be calibrated by a Hölder modulus that is large and unreliable on real data. DTAP does not enlarge the class; it exploits the finer convex/pseudoconvex split *inside* it to recover a quantitative gradient inequality on part of the system, and with it a step that needs no modulus at all.

[Former Section 2.3] Optimization-based feature selection

Two embedded, optimization-based models anchor our application (see the surveys [16, 24, 30]). Wang et al. [32] pose selection as a fractional program (a redundancy

quadratic over a relevance linear form on the simplex, pseudoconvex there), solved by a projection neural network. Wang et al. [33] propose AFRSFCV, a fuzzy-rough model injecting class variance into the similarity relation and minimizing a softmax loss that combines a lower-approximation fit, a decision-consistency term, and Tikhonov regularization. Each ships its own solver and objective; neither combines criteria, and optimizing either alone tends to yield overly sparse selections. Our framework reuses exactly these objectives as feasibility constraints, retaining their semantics while unifying them.

[Former Section 2.4] Decomposition and the positioning of dtap

Decomposing a heterogeneous problem into homogeneous blocks updated by block-appropriate rules is a recurring theme [3, 9]. DTAP applies this at the level of *constraints*: a residual-adaptive step where a quantitative gradient inequality holds (convex track), and a safeguarded step where only sign information is available (pseudoconvex track). A third strand pursues strong convergence via Halpern, viscosity, or inertial modifications [9, 34], but typically for linear SFP or a single averaged operator; DTAP reaches it by constructing a tailored one-step quasi-Fejér inequality (following the general framework of Combettes 10) plus a pseudoconvex sign argument, needing neither the exact Lipschitz constant nor the projection onto the complex feasible set.

Table 1 makes the positioning explicit: no single column of it isolates DTAP, because each ingredient of the method already has precedent. A constant-free step appears in [25] and, in diminishing form, in [37]; the same residual-adaptive formula is reused verbatim by [8], though that method additionally needs the S-subdifferential modulus and so is not itself constant-free. Projecting only onto the base set is done by both [19] and [37]; whole-sequence convergence is proved in [23, 37]; and a max-type scalarization driven by the Clarke subdifferential on an active set is the device of [37]. The contribution of DTAP lies in the conjunction of two elements not previously combined: the admissible *class* and the *step that drives it*. On the class, the two nearest neighbours sit on opposite sides: [19] lies strictly above ours (quasi-convex), while [37] lies beside it (quasilinear, which neither contains nor is contained in convex-plus-pseudoconvex). Consequently neither subsumes the problem we solve. On the step, the residual-adaptive rule (3) has always been licensed by a quantitative inequality that its setting supplies for free: firm nonexpansiveness of $I - P_Q$ under a linear $\mathcal{A}$ in [8, 25], and co-coercivity of ∇p under a convex proximity function in [23]. We obtain the same inequality, $\langle g^{\mathrm{cvx}}, x - x^\star \rangle \geq 2\Phi^{\mathrm{cvx}}$, from convexity of the individual components, which persists under both a nonlinear F and the presence of pseudoconvex components elsewhere in the system; those components are then absorbed on a second track that uses only the sign pseudoconvexity still guarantees. This inequality lets the convex part keep a non-diminishing, residual-scaled step, whereas [37] must fall back on a prescribed diminishing sequence. To our knowledge DTAP is therefore the first method for the normal-box NSFP with *mixed convex–pseudoconvex* components whose step is closed-form and free of every constant (operator norm, Lipschitz constant, Hölder modulus, S-subdifferential modulus), which requires neither linearity of the map nor convexity of the aggregate proximity function nor two-sided quasilinearity of the components, and which converges strongly to a feasible point.

2. Preliminaries

Throughout, $H = \mathbb{R}^n$ is Euclidean space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$. All convergence statements concern the norm topology. Bounded sequences have

Former Table 1. Positioning of DTAP against twelve representative prior works. Two columns carry ✓/×: 'Const.-free' marks methods whose iteration needs no prescribed constant: no operator norm, no Lipschitz or Hölder modulus, and no S-subdifferential modulus; 'Only P_K' marks methods that project only onto the base set, with no projection onto the output set or the solution set. 'Step-size rule' gives the update in closed form, and P_Ω in it denotes a projection onto the base restriction set of the multiple-sets model. No single column isolates DTAP: the 'Const.-free' tick is shared with [23, 25, 36, 37] and the 'Only P_K' tick with [19, 32, 37], so in particular Yen et al. carry both. DTAP is set apart only by the conjunction of the last two columns, specifically a constant-free, residual-adaptive step on a *mixed convex–pseudoconvex* class. On that class the two nearest neighbours lie on opposite sides: the quasi-convex row of [19] is strictly *broader* (it contains convex-plus-pseudoconvex, but treats it uniformly and pays with a Hölder modulus), whereas the quasilinear row of [37] is *incomparable* (it neither contains nor is contained in our class), so neither subsumes the problem DTAP solves.

Work	**Problem class**	**Map**	**Output set**	**Step-size rule**	**Const.-free**	**Only P_K**	**Constraint**
Censor–Elfving [6]	Split feas. (origin)	Lin.	General convex	Bregman multiprojection	×	×	convex
Byrne [5] (CQ)	Split feasibility	Lin.	General convex	$\tau < 2/\|\mathcal{A}\|^2$ fixed	×	×	convex
Fukushima [15]	Variational inequality	Lin.	Level set of cvx c	Relaxed (half-space)	×	×	convex
Yang [36] (relaxed CQ)	Split feasibility	Lin.	Level set of cvx c	$\rho_n f/\|\nabla f\|^2$, half-sp.	✓	×	convex
López et al. [25]	Split feasibility	Lin.	General convex	$\rho_n f/\|\nabla f\|^2$, $\rho_n \in (0,4)$	✓	×	convex
Xu [34]	Split feas. (Hilbert)	Lin.	General convex	KM / regularized	×	×	convex
Li–Han–Zhang [23]	Nonlinear MSFP	Non-lin.	$\bigcap_j$ convex	β_k backtracked (needs p cvx)	✓	×	convex (p cvx)
Yen et al. [37]	Nonlin. SFP (Nash)	Non-lin.	Convex	$\beta_k/\max\{\delta_k, \|g_k\|\}$ dimin.	✓	✓	quasilinear
Chen et al. [8]	Nonconvex SFP	Lin.	S-subdiff. level sets	$\lambda_k f_k/\|\nabla f_k\|^2$, $\lambda_k \in (0,2)$	×	×	S-subdiff. noncvx
Hu et al. [19]	Quasicvx feasibility	Non-lin.	Inequality system	Const./dynamic subgrad. (Hölder)	×	✓	quasi-cvx
Wang et al. [32]	Frac.-prog. selection	Non-lin.	Simplex	Neurodynamic (PNN)	×	✓	pseudo-cvx
Wang et al. [33]	Fuzzy-rough selection	Non-lin.	$\mathbb{R}^m$ (softmax)	Gradient descent	×	×	nonconvex
This work	**Normal-box NSFP**	**Non-lin.**	**Normal box**	$\lambda\Phi^{\mathrm{cvx}}/\|g^{\mathrm{cvx}}\|^2 \oplus \gamma_k$	✓	✓	**cvx + pseudo**

convergent subsequences. The standard projection and generalized convexity facts below can be found in [2, 27].

Throughout, $H := \mathbb{R}^n$ denotes the n-dimensional Euclidean space with inner product $\langle x, y\rangle := \sum_{i=1}^n x_i y_i$ and induced norm $\|x\| := \sqrt{\langle x, x\rangle}$. Every bounded subset of H is relatively compact, so by Bolzano–Weierstrass every bounded sequence has a convergent subsequence; convergence is in norm, $x^k \to x$ iff $\|x^k - x\| \to 0$. Most statements below are standard [1, 2, 27]; the two results invoked directly in the convergence analysis (nonexpansiveness of the projection and a deterministic quasi-Fejér lemma) are proved in Section 3.

We collect the main notation here. The ambient space is $H = \mathbb{R}^n$ with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$; K is the base closed convex set (the simplex Δ_{p-1} in the application) and P_K its metric projection. The component map $F = (F_1, \dots, F_m)$, threshold vector b, and normal box $Q = \prod_i (-\infty, b_i]$ define the feasible set $\Omega = \{x \in K : F_i(x) \le b_i \ \forall i\}$; the Tchebyshev error is $J(x) = \max_i (F_i(x) - b_i)^+$ and the componentwise violation $v_i(x) = (F_i(x) - b_i)^+$. The constraint indices split into convex and pseudoconvex sets $I^{\mathrm{cvx}}, I^{\mathrm{ps}}$, with per-track maximal violations $J^{\mathrm{cvx}}, J^{\mathrm{ps}}$, δ-approximate active sets $K^{\mathrm{cvx}}(x), K^{\mathrm{ps}}(x)$, search directions $g^{\mathrm{cvx}}, g^{\mathrm{ps}}$, step sizes $\eta_k^{\mathrm{cvx}}, \eta_k^{\mathrm{ps}}$, soft weight μ_k, and Polyak numerator $\Phi^{\mathrm{cvx}}(x)$; the algorithm parameters are the Polyak relaxation $\rho_{\mathrm{cvx}} \in (0, 2)$, the safeguarded sequence $\{\gamma_k\}$ with $\sum \gamma_k = \infty$ and $\sum \gamma_k^2 < \infty$, and the active-set width $\delta > 0$. In the application, n, p, r are the numbers of samples, features, and classes, $w \in \Delta_{p-1}$ the feature-weight vector, ρ and Q the Fisher-score vector and redundancy matrix, $\underline{R_k}(D)$, P, Y the lower-approximation, decision, and label matrices, and θ the cumulative selection threshold.

[**Former Section 3.1**] **Euclidean projection**

Definition 3.1 (Metric projection). Let $K \subseteq H$ be nonempty closed convex. For $x \in H$, a point $p \in K$ with $\|p - x\| = \inf_{y \in K} \|y - x\|$ is the *projection* of x onto K, written $p = P_K(x)$. The map $P_K : H \to K$ is the *metric projection* onto K.

For closed convex K, $P_K(x)$ exists and is unique for every $x \in H$ [2], and is characterized variationally as follows.

Proposition 3.2 (Variational characterization). Let $K \subseteq H$ be nonempty closed convex. For $x \in H$ and $p \in K$, one has $p = P_K(x)$ if and only if

$$\langle x - p, y - p\rangle \le 0 \qquad \text{for all } y \in K.$$

Proposition 2.1 underlies the classical regularity properties of the projection: nonexpansiveness, firm nonexpansiveness, and inverse strong monotonicity [2]. Nonexpansiveness is used repeatedly in Section 3 and is isolated there as Lemma 3.2. The simplex projection, needed for the feature-selection application, is computable in $O(m \log m)$ time [11] and is treated as a black box.

For a nonempty closed convex set $K \subseteq H$, the metric projection $P_K(x)$ is the unique minimizer of $\|y - x\|$ over $y \in K$.

Proposition 2.1. *For $x \in H$ and $p \in K$, $p = P_K(x)$ if and only if*

$$\langle x - p, y - p\rangle \le 0 \qquad \text{for every } y \in K. \tag{4}$$

This characterization gives the projection inequality used in Lemma 3.2. The sort-based projection onto a p-dimensional simplex takes $O(p \log p)$ operations [11].

[**Former Section 3.2**] **Convex and pseudoconvex functions**

Let K be a nonempty convex subset of $H = \mathbb{R}^n$.

Definition 3.3 (Convex function). $f : K \to \mathbb{R}$ is *convex on* K if for all $x, y \in K$ and $\lambda \in [0,1]$, $f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda) f(y)$.

The next notion, introduced by Mangasarian [26], relaxes convexity while preserving the property that a stationary point is a global minimizer.

Definition 3.4 (Pseudoconvex function). Let $f : K \to \mathbb{R}$ be differentiable on an open neighbourhood of K. Then f is *pseudoconvex on* K if for all $x, y \in K$, $\langle \nabla f(x), y - x \rangle \ge 0 \Rightarrow f(y) \ge f(x)$.

Definition 2.2 is equivalent to its contrapositive: for all $x, y \in K$,

$$f(y) < f(x) \implies \langle \nabla f(x), y - x \rangle < 0.$$

Form (5) is the main tool of the pseudoconvex-track analysis in Section 3: it converts a strict decrease in function value into a strict sign condition on the directional derivative.

A function $f : K \to \mathbb{R}$ is convex if $f((1-t)x + ty) \le (1-t)f(x) + tf(y)$ for $x, y \in K$ and $t \in [0,1]$. The following weaker derivative condition was introduced by Mangasarian [26].

Definition 2.2. A function differentiable on an open neighbourhood of K is pseudoconvex on K if, for every $x, y \in K$,

$$f(y) < f(x) \quad \Longrightarrow \quad \langle \nabla f(x),\, y - x \rangle < 0. \tag{5}$$

Every differentiable convex function is pseudoconvex. Pseudoconvex functions have convex sublevel sets, although the functions themselves need not be convex. This distinction is relevant to the mixed constraint functions studied below.

[**Former Section 3.3**] **Feature selection**

Feature selection is a preprocessing step that compresses an initial feature set into a subset retaining information relevant to the target while discarding redundant and irrelevant features [16, 30].

Definition 3.5 (Feature selection). Let $U = \{x_1, \dots, x_n\}$ be n samples and $A = \{a_1, \dots, a_p\}$ a set of p features, with data matrix $X \in \mathbb{R}^{n \times p}$ whose k-th column holds the values of a_k. *Feature selection* seeks $A^\star \subseteq A$ preserving the discriminative information of A for a given learning target while $|A^\star| \ll p$.

Following [30], methods are categorized by label usage (supervised, unsupervised, semi-supervised); by interaction with the learner (filter, wrapper, embedded); and by output form (ranking, subset, weighting). This paper concerns *supervised, embedded, weighting* methods: those that select features by solving a constrained continuous optimization problem whose output is a weight vector $w \in \mathbb{R}^p$ of relative feature importances. This is the class that can be recast as finding a common point of two constraint systems on two spaces (one expressing admissibility of the weight vector, the other effectiveness of the selected set), which is the structure of the split feasibility problem treated next.

[**Former Section 3.4**] **Split feasibility and its nonlinear extension**

Byrne's *CQ* recursion (2) is Fejér-monotone with respect to the solution set but requires $\|\mathcal{A}\|$ and P_Q. The relaxed methods of [15, 36] remove P_Q by projecting onto an outer half-space, and the self-adaptive rule of [25] removes $\|\mathcal{A}\|$ by calibrating the step from the residual; we inherit the latter, constant-free philosophy. Replacing the linear $\mathcal{A}$

by a nonlinear F yields NSFP (3), which raises four specific challenges that our model and algorithm meet.

(D1) *Loss of the global gradient inequality.* When some F_i are nonconvex, the supporting-hyperplane inequality fails globally, so hyperplanes built from ∇F_i may cut off the solution $x^\star$ and invalidate CQ-type convergence.

(D2) *Choice of the residual on the output set.* For a *normal box* $Q = (-\infty, b_1] \times \cdots \times (-\infty, b_m]$ [18] both the Euclidean and the sup-norm residuals are available in closed form, the former as $d_2(F(x), Q)^2 = \sum_i v_i(x)^2$ and the latter as $d_\infty(F(x), Q) = \max_i v_i(x)$ with $v_i(x) = (F_i(x) - b_i)^+$. The Euclidean choice is the proximity function of the multiple-sets solver [23], which aggregates all violations into one smooth surrogate; that aggregation, however, forces a uniform step on the whole system. The sup-norm residual instead certifies feasibility constraint by constraint ($J(x) = 0$ iff every $v_i(x) = 0$), and its maximizers single out the binding constraints, which is the information a per-constraint step policy requires. We therefore adopt the Tchebyshev scalarization for its structure, which matches the dual-track design of Section 3.2; the choice is not forced by closed-form availability, since the Euclidean residual has one too.

(D3) *Nonsmoothness of the max-type residual.* The Tchebyshev residual is a maximum of componentwise violations, nondifferentiable where several constraints are simultaneously active; the full Clarke subdifferential is delicate when the active set changes between iterations.

(D4) *Dependence on a Lipschitz constant.* Methods for NSFP typically need a step tied to the Lipschitz constant of ∇F, hard to estimate; an adaptive, constant-free step is preferable.

To address (D1)–(D4) together, we add the structural hypothesis that each F_i is convex or pseudoconvex, adopt the constraint-certifying Tchebyshev residual of (D2), treat (D3) through an approximate active set, and design a Lipschitz-free adaptive step for (D4).

3. The ~~dtap~~ DTAP algorithm and its convergence analysis

We formulate the normal-box NSFP, state DTAP, and prove convergence of the whole sequence to a feasible point.

Building on the preliminaries of Section 2, we now formulate the normal-box NSFP and its Tchebyshev scalarization, decompose the constraint system into a convex and a pseudoconvex track, fix the standing assumptions, and state the DTAP algorithm; we then establish its strong convergence to a feasible point.

3.1. Problem statement and Tchebyshev scalarization

Let ~~$H := \mathbb{R}^n$, let~~ $K \subseteq H$ be nonempty closed convex, ~~fix~~ and let $U \subseteq H$ be an open neighbourhood of K. Fix an integer ~~$m \geq 1$~~$m \geq 1$, let ~~$F = (F_1, \ldots, F_m) : H \to \mathbb{R}^m$~~$F = (F_1, \ldots, F_m) : U \to \mathbb{R}^m$, let $b = (b_1, \ldots, b_m) \in \mathbb{R}^m$, and set the normal box $Q := (-\infty, b_1] \times \cdots \times (-\infty, b_m]$. We study the instance of (3)

$$\text{find } x^\star \in \Omega, \qquad \Omega := \big\{x \in K : F_i(x) \leq b_i,\ i = 1, \ldots, m\big\}, \tag{6}$$

and call Ω the *feasible set*. To turn $F(x) \in Q$ into a tractable merit function, we measure the residual in the sup-norm. For $z = F(x)$, $d_\infty(z, Q) = \inf_{q \le b} \max_{1 \le i \le m} |z_i - q_i|$. By the box structure the infimum separates over coordinates: if $z_i \le b_i$ then $\inf_{q_i \le b_i} |z_i - q_i| = 0$ (at $q_i = z_i$), and if $z_i > b_i$ then it equals $z_i - b_i$ (at $q_i = b_i$). Hence $\inf_{q_i \le b_i} |z_i - q_i| = (z_i - b_i)^+$ with $(t)^+ := \max\{t, 0\}$, so

$$d_\infty(z, Q) = \max_{1 \le i \le m} (z_i - b_i)^+. \tag{7}$$

Define the *error function*

$$J(x) := d_\infty\big(F(x), Q\big) = \max_{1 \le i \le m} \big(F_i(x) - b_i\big)^+. \tag{8}$$

Then $F(x) \in Q$ iff $J(x) = 0$, so finding a feasible point amounts to minimizing J over K. ~~The nonsmoothness of J (difficulty (D3)) is handled in~~ Formula (8) gives a closed-form feasibility residual whose maximizers identify the most violated constraints. Section 3.2 ~~by~~ introduces an approximate active set ~~. Formula (8) settles (D2): computed for the price of m scalar evaluations, J vanishes exactly at feasible points and its maximizers pick out the binding constraints, so it is at once a merit function and the selector the dual-track step will read~~to handle the nonsmooth maximum.

3.2. Decomposition and dual-track scalarization

~~A common strategy in large-scale optimization is not to solve a heterogeneous problem with one uniform technique, but to decompose it into structurally homogeneous blocks and update each with the rule best suited to it [3]. In large-scale sparse regression, for instance, the objective splits into a smooth data-fidelity part and a nonsmooth regularizer, each handled by its own update; the success of such schemes comes from exploiting the structure of each block rather than imposing a common treatment.~~

~~Problem (6) has an analogously heterogeneous constraint system: most constraints have simple geometry (convex, even affine), while a few are more complex: nonconvex but pseudoconvex. Folding all violations into one error function J and treating them uniformly would force the safeguarded diminishing step on the whole system, wasting the structure of the convex block and slowing convergence. Following the decomposition philosophy, we split the constraint index set into a *convex track* and a *pseudoconvex track* and give each its own step rule: a Polyak step [29] on the convex track and a safeguarded diminishing step on the pseudoconvex track. The two are reconciled by a soft proportional weight.~~

We split the constraint index set into a *convex track* and a *pseudoconvex track*. The convex track uses a Polyak step [29]; the pseudoconvex track uses a safeguarded diminishing step. A proportional weight combines the two updates.

Fix a parameter $\delta > 0$. For ~~$x \in H$~~ $x \in U$ define the componentwise violations and the maximal violation

$$v_i(x) := \big(F_i(x) - b_i\big)^+, \qquad J(x) = \max_{1 \le i \le m} v_i(x). \tag{9}$$

Thus $x \in \Omega$ iff $x \in K$ and $J(x) = 0$. Partition $I := \{1, \dots, m\}$ into disjoint groups $I = I^{\mathrm{cvx}} \cup I^{\mathrm{ps}}$, where I^{cvx} indexes the convex constraints and I^{ps} the pseudoconvex ones

(made precise in Assumption 1). The *per-track maximal violations* are

$$J^{\mathrm{cvx}}(x) := \max_{i \in I^{\mathrm{cvx}}} v_i(x) \quad (=0 \text{ if } I^{\mathrm{cvx}} = \emptyset), \tag{10}$$

$$J^{\mathrm{ps}}(x) := \max_{j \in I^{\mathrm{ps}}} v_j(x) \quad (=0 \text{ if } I^{\mathrm{ps}} = \emptyset), \tag{11}$$

so $J(x) = \max\{J^{\mathrm{cvx}}(x), J^{\mathrm{ps}}(x)\}$. To cope with the nonsmoothness of the maximum~~(D3)~~, we replace the exact maximizers by a δ-enlarged *approximate active set* on each track:

$$K^{\mathrm{cvx}}(x) := \begin{cases} \{i \in I^{\mathrm{cvx}} : v_i(x) \geq J^{\mathrm{cvx}}(x) - \delta\} & \text{if } J^{\mathrm{cvx}}(x) > 0, \\ \emptyset & \text{if } J^{\mathrm{cvx}}(x) = 0, \end{cases} \tag{12}$$

$$K^{\mathrm{ps}}(x) := \begin{cases} \{j \in I^{\mathrm{ps}} : v_j(x) \geq J^{\mathrm{ps}}(x) - \delta\} & \text{if } J^{\mathrm{ps}}(x) > 0, \\ \emptyset & \text{if } J^{\mathrm{ps}}(x) = 0. \end{cases} \tag{13}$$

On these sets we form the *per-track local functions and directions*

$$\Phi^{\mathrm{cvx}}(x) := \tfrac{1}{2} \sum_{i \in K^{\mathrm{cvx}}(x)} v_i(x)^2, \qquad g^{\mathrm{cvx}}(x) := \sum_{i \in K^{\mathrm{cvx}}(x)} v_i(x) \nabla F_i(x), \tag{14}$$

$$\Phi^{\mathrm{ps}}(x) := \tfrac{1}{2} \sum_{j \in K^{\mathrm{ps}}(x)} v_j(x)^2, \qquad g^{\mathrm{ps}}(x) := \sum_{j \in K^{\mathrm{ps}}(x)} v_j(x) \nabla F_j(x). \tag{15}$$

The vectors $g^{\mathrm{cvx}}(x), g^{\mathrm{ps}}(x)$ are the two search directions, and $\Phi^{\mathrm{cvx}}(x)$ supplies the Polyak numerator on the convex track. Along the iterates we abbreviate $J_k^{\mathrm{cvx}} := J^{\mathrm{cvx}}(x^k)$, $g_k^{\mathrm{cvx}} := g^{\mathrm{cvx}}(x^k)$, $\Phi_k^{\mathrm{cvx}} := \Phi^{\mathrm{cvx}}(x^k)$, and likewise on the pseudoconvex track.

~~[Former Section 4.3] Standing assumptions~~

3.3. Assumptions

Assumption 1. We assume (A1)–(A3):

(A1) (a) $\Omega \neq \emptyset$. (b) $I = \{1, \dots, m\}$ is partitioned as $I = I^{\mathrm{cvx}} \cup I^{\mathrm{ps}}$; for each i, ~~$F_i : H \to \mathbb{R}$~~$F_i : U \to \mathbb{R}$, and the restriction $F_i|_K$ is convex if $i \in I^{\mathrm{cvx}}$ and pseudoconvex if $i \in I^{\mathrm{ps}}$.

(A2) Each ~~$F_i : H \to \mathbb{R}$ is~~ $F_i : U \to \mathbb{R}$ is continuously differentiable on ~~H, and ∇F_i is continuous on every compact subset of H~~U.

(A3) (i) $\delta > 0$ fixed. (ii) $\rho_{\mathrm{cvx}} \in (0, 2)$. (iii) $\{\gamma_k\} \subset (0, \infty)$ with $\sum_{k=0}^{\infty} \gamma_k^2 < \infty$ and $\sum_{k=0}^{\infty} \gamma_k = \infty$.

The open-neighbourhood formulation permits fractional functions whose denominators are positive on K; the algorithm evaluates the functions and gradients only at points of K. Although some component functions may be nonconvex, Ω is convex: differentiable pseudoconvex functions are quasiconvex on K, so all constraint sublevel sets in K are convex.

Example 3.1. The sequence $\gamma_k := 1/(k+1)$ satisfies (A3)(iii), since $\sum_{k \geq 0} (k+1)^{-2} < \infty$ while $\sum_{k \geq 0} (k+1)^{-1} = \infty$. More generally $\gamma_k = (k+c)^{-s}$ with $c > 0$ and $s \in (1/2, 1]$ qualifies; the experiments use $s = 0.55$.

3.4. The ~~dtap~~ DTAP algorithm

The Dual-Track Adaptive Projection algorithm combines the two track-specific step sizes through the soft weight μ_k and projects the combined update onto the base convex set K. It is stated as Algorithm 1.

Algorithm 1 Dual-Track Adaptive Projection (~~DTAP~~DTAP) for the normal-box NSFP (6).

Input: map $F = (F_1, \dots, F_m)$, thresholds b, base set K, index split $I^{\mathrm{cvx}} \cup I^{\mathrm{ps}}$.
Parameters: $\delta > 0$, tolerance $\varepsilon > 0$, $\rho_{\mathrm{cvx}} \in (0, 2)$, sequence $\{\gamma_k\}$ satisfying (A3)(iii).
Initialization: choose $x^0 \in K$; set $k \leftarrow 0$.

Step 1 (per-track quantities).

(1a) Compute $v_i(x^k)$ for all i, and $J_k^{\mathrm{cvx}} = \max_{i \in I^{\mathrm{cvx}}} v_i(x^k)$, $J_k^{\mathrm{ps}} = \max_{j \in I^{\mathrm{ps}}} v_j(x^k)$ (each 0 if its track is empty).

(1b) Form the approximate active sets $K_k^{\mathrm{cvx}}, K_k^{\mathrm{ps}}$ via (12)–(13).

(1c) Form Φ_k^{cvx}, g_k^{cvx}, g_k^{ps} via (14)–(15).

(1d) Per-track step sizes (Polyak on the convex track, safeguarded on the pseudoconvex track):

$$\eta_k^{\mathrm{cvx}} = \begin{cases} \rho_{\mathrm{cvx}} \dfrac{2\Phi_k^{\mathrm{cvx}}}{\|g_k^{\mathrm{cvx}}\|^2} & \text{if } \|g_k^{\mathrm{cvx}}\| > 0, \\ 0 & \text{otherwise,} \end{cases} \qquad \eta_k^{\mathrm{ps}} = \frac{\gamma_k}{\max\{1, \|g_k^{\mathrm{ps}}\|\}}.$$

(1e) Soft proportional weight: $\mu_k = \dfrac{J_k^{\mathrm{cvx}}}{J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}}}$ if $J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}} > 0$, and $\mu_k = 1/2$ otherwise.

(1f) Combine the two directions: $y^k = x^k - [\mu_k \eta_k^{\mathrm{cvx}} g_k^{\mathrm{cvx}} + (1 - \mu_k)\eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}]$.

Step 2 (projection). $x^{k+1} = P_K(y^k)$.

Step 3 (stopping). If $J(x^k) \le \varepsilon$, stop and return x^k; otherwise set $k \leftarrow k + 1$ and go to Step 1.

Remark 1. The proportional weight is an adaptive allocation based on the two maximal violations; we do not claim that it is the only convergent choice or that it is optimal. Its specific form is used in (27) to control the convex-track residual and in Claim 2 of Stage 2 of Theorem 3.12 to ensure that a persistently violated pseudoconvex track retains positive weight. Although the one-step inequality holds for any $\mu_k \in [0, 1]$, this alone does not establish convergence for every such sequence.

Remark 2. The iteration does not require prescribed operator norms, Lipschitz constants, or Hölder moduli. The proof nevertheless uses finite bounds on function values and gradients on a compact set containing the iterates, after boundedness has been proved. Bounded gradients imply Lipschitz continuity of the functions on that convex compact set, but these bounds are not algorithm inputs. Assumption (A2) does not imply Lipschitz continuity of the gradients, and the theorem does not require it. The parameters ρ_{cvx}, δ, and $\{\gamma_k\}$ remain part of the algorithm.

For the analysis we write the combined increment as

$$d_k := \mu_k \eta_k^{\mathrm{cvx}} g_k^{\mathrm{cvx}} + (1 - \mu_k)\eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}, \tag{16}$$

so $y^k = x^k - d_k$ and $x^{k+1} = P_K(x^k - d_k)$.

3.5. Per-iteration cost

Each DTAP iteration performs a fixed set of F-agnostic operations: it evaluates the componentwise violations $v_i(x^k)$ and the active-set gradients $\{\nabla F_i(x^k) : i \in K_k^{\mathrm{cvx}} \cup K_k^{\mathrm{ps}}\}$, forms the two search directions $g_k^{\mathrm{cvx}}, g_k^{\mathrm{ps}}$ as weighted sums of these gradients, computes the two scalar step sizes $\eta_k^{\mathrm{cvx}}, \eta_k^{\mathrm{ps}}$, and closes with a single projection P_K onto the base set. Unlike Byrne's CQ recursion (2), there is no power iteration to estimate an operator norm $\|\mathcal{A}\|$ and no performs one projection onto K. It uses neither an operator-norm estimate nor a line search; the . The cost is thus dominated by the constraint evaluations and the one projection. The concrete order cost for the feature-selection instance , which depends on the explicit form of the F_i, is deferred to is given in Section 4.1 (Remark 6).

The convergence analysis of DTAP is more involved than for classical CQ schemes because the two constraint tracks are asymmetric: on the convex track the gradient inequality gives a quantitative lower bound of order $v_i(x)^2$, hence a strict decrease, whereas on the pseudoconvex track the contrapositive of pseudoconvexity gives only sign information. The two are reconciled in a single one-step quasi-Fejér inequality with a carefully chosen Young parameter, and a contradiction argument rules out cluster points outside the feasible set. We then obtain the main result, strong convergence to a feasible point (Theorem 3.12), and deduce the two degenerate structural regimes as corollaries.

Throughout, $\{x^k\}$ is the infinite sequence generated by Algorithm 1 with the stopping test removed(i.e. $\varepsilon = 0$ in Step 3). As shown below (Lemma 3.10 and Theorem 3.12), $J(x^k) \to 0$; hence for any $\varepsilon > 0$ there is a finite k with $J(x^k) \le \varepsilon$, so the finite-stopping version terminates after finitely many steps with an ε-feasible point.

3.6. Auxiliary lemmas

We begin with two preparatory results used repeatedly below.

Lemma 3.2. *Let $K \subseteq H$ be nonempty closed convex. For every $y \in H$ and $z \in K$,*

$$\|P_K(y) - z\|^2 \le \|y - z\|^2 - \|y - P_K(y)\|^2 \le \|y - z\|^2 . \tag{17}$$

Proof. Put $p := P_K(y)$. By Proposition 2.1, $\langle y - p, z - p\rangle \le 0$, so expanding $\|y - z\|^2 = \|y - p\|^2 + 2\langle y - p, p - z\rangle + \|p - z\|^2$ and using $\langle y - p, p - z\rangle \ge 0$ gives $\|y - z\|^2 \ge \|y - p\|^2 + \|p - z\|^2$, which is (17). □

Lemma 3.3. *Let $\{a_k\} \subset [0, \infty)$ and $\{\varepsilon_k\} \subset [0, \infty)$ satisfy $a_{k+1} \le a_k + \varepsilon_k$ for all k and $\sum_{k=0}^{\infty} \varepsilon_k < \infty$. Then $\{a_k\}$ converges to a finite limit.*

Proof. $b_k := a_k - \sum_{l<k} \varepsilon_l$ is nonincreasing (from $a_{k+1} \le a_k + \varepsilon_k$) and bounded below by $-\sum_l \varepsilon_l > -\infty$, hence converges; since $\sum_{l<k} \varepsilon_l \to S < \infty$, $a_k = b_k + \sum_{l<k} \varepsilon_l$ converges. □

On the convex track, convexity gives a quantitative link between the gradient and the violation.

Lemma 3.4. *For $x^\star \in \Omega$, $i \in I^{\mathrm{cvx}}$, and $x \in K$,*

$$v_i(x)\ \langle \nabla F_i(x),\, x - x^\star\rangle \ge v_i(x)^2.$$

Proof. If $v_i(x) = 0$ both sides vanish. Let $v_i(x) > 0$, i.e. $F_i(x) > b_i$. Define $\varphi : [0,1] \to \mathbb{R}$ by $\varphi(t) := F_i((1-t)x + tx^\star)$; since K is convex, $[x, x^\star] \subset K$, so φ is well defined, and by (A1)(b) it is convex, while by (A2) it is differentiable with $\varphi'(0^+) = \langle \nabla F_i(x),\, x^\star - x\rangle$. The one-dimensional gradient inequality gives $\varphi(1) \geq \varphi(0) + \varphi'(0^+)$, i.e. $F_i(x^\star) \geq F_i(x) + \langle \nabla F_i(x),\, x^\star - x\rangle$. As $x^\star \in \Omega$ we have $F_i(x^\star) \leq b_i < F_i(x)$, whence $\langle \nabla F_i(x),\, x - x^\star\rangle \geq F_i(x) - b_i = v_i(x)$. Multiplying by $v_i(x) > 0$ gives the claim. □

Remark 3. Summing Lemma 3.4 over the approximate active set K_k^{cvx} yields $\langle g_k^{\mathrm{cvx}},\, x^k - x^\star\rangle \geq \sum_{i \in K_k^{\mathrm{cvx}}} v_i(x^k)^2 = 2\Phi_k^{\mathrm{cvx}}$, a bound used in the proofs of Lemmas 3.8 and 3.10.

On the pseudoconvex track, only a sign is available.

Lemma 3.5. *For $x^\star \in \Omega$, $j \in I^{\mathrm{ps}}$, and $x \in K$, $v_j(x)\, \langle \nabla F_j(x),\, x - x^\star\rangle \geq 0$, with strict inequality when $v_j(x) > 0$.*

Proof. If $v_j(x) = 0$ both sides vanish. If $v_j(x) > 0$ then $F_j(x) > b_j \geq F_j(x^\star)$. By (A1)(b), $F_j|_K$ is pseudoconvex; applying the contrapositive (5) with $y = x^\star$ gives $\langle \nabla F_j(x),\, x^\star - x\rangle < 0$, i.e. $\langle \nabla F_j(x),\, x - x^\star\rangle > 0$. Multiplying by $v_j(x) > 0$ yields the strict inequality. □

Finally, two compactness-based norm bounds on the search directions.

Lemma 3.6. *Assume $I^{\mathrm{cvx}} \neq \emptyset$ and let ~~$B \subset H$~~ $B \subset U$ be compact. With $L_B := \max_{i \in I^{\mathrm{cvx}}} \sup_{x \in B} \|\nabla F_i(x)\|$ (finite by (A2)), for every $x \in B$,*

$$\|g^{\mathrm{cvx}}(x)\|^2 \leq 2\, |I^{\mathrm{cvx}}|\, L_B^2\, \Phi^{\mathrm{cvx}}(x).$$

Proof. Put $K' = K^{\mathrm{cvx}}(x)$, $m' = |K'| \leq |I^{\mathrm{cvx}}|$. By the triangle and Cauchy–Schwarz inequalities,

$$\|g^{\mathrm{cvx}}(x)\| = \Big\| \sum_{i \in K'} v_i(x) \nabla F_i(x) \Big\| \leq \Big(\sum_{i \in K'} v_i(x)^2 \Big)^{1/2} \Big(\sum_{i \in K'} \|\nabla F_i(x)\|^2 \Big)^{1/2}.$$

Since $\sum_{i \in K'} v_i(x)^2 = 2\Phi^{\mathrm{cvx}}(x)$ and $\sum_{i \in K'} \|\nabla F_i(x)\|^2 \leq m' L_B^2 \leq |I^{\mathrm{cvx}}|\, L_B^2$, squaring gives the claim. □

Lemma 3.7. *Assume $I^{\mathrm{ps}} \neq \emptyset$ and let ~~$B \subset H$~~ $B \subset U$ be compact. With $L'_B := \max_{j \in I^{\mathrm{ps}}} \sup_{x \in B} \|\nabla F_j(x)\|$ and $V_B := \max_{j \in I^{\mathrm{ps}}} \sup_{x \in B} v_j(x)$ (both finite by (A2)), for every $x \in B$,*

$$\|g^{\mathrm{ps}}(x)\| \leq |I^{\mathrm{ps}}|\, V_B\, L'_B.$$

Proof. By the triangle inequality, $\|g^{\mathrm{ps}}(x)\| = \big\|\sum_{j \in K^{\mathrm{ps}}(x)} v_j(x) \nabla F_j(x)\big\| \leq \sum_{j \in K^{\mathrm{ps}}(x)} v_j(x)\, \|\nabla F_j(x)\| \leq |K^{\mathrm{ps}}(x)|\, V_B L'_B \leq |I^{\mathrm{ps}}|\, V_B L'_B$. □

3.7. Boundedness and decay of the iteration error

The ~~central result is a novel one-step~~ following quasi-Fejér inequality~~(conceptually rooted in~~, within the framework of~~Combettes 10), from which we extract~~ Combettes [10], yields boundedness of ~~$\{x_k\}$~~ $\{x^k\}$ and decay of the convex-track violation. Set

$$c_0 := \frac{\rho_{\mathrm{cvx}}(2-\rho_{\mathrm{cvx}})}{2} \in \left(0, \tfrac{1}{2}\right], \qquad C_0 := \frac{2+\rho_{\mathrm{cvx}}}{2-\rho_{\mathrm{cvx}}} \in (1, \infty), \tag{18}$$

and

$$T_k := \begin{cases} \dfrac{(2\Phi_k^{\mathrm{cvx}})^2}{\left\|g_k^{\mathrm{cvx}}\right\|^2} & \text{if } \left\|g_k^{\mathrm{cvx}}\right\| > 0, \\ 0 & \text{if } \left\|g_k^{\mathrm{cvx}}\right\| = 0. \end{cases} \tag{19}$$

For fixed $x^\star \in \Omega$ write $R_k^{x^\star} := \left\langle g_k^{\mathrm{ps}},\, x^k - x^\star \right\rangle$.

Lemma 3.8. *For every $x^\star \in \Omega$ and $k \geq 0$,*

$$\left\|x^{k+1} - x^\star\right\|^2 \leq \left\|x^k - x^\star\right\|^2 - c_0\mu_k T_k - 2(1-\mu_k)\eta_k^{\mathrm{ps}} R_k^{x^\star} + C_0\gamma_k^2. \tag{20}$$

Proof. Apply Lemma 3.2 with $y = y^k$ and $z = x^\star \in K$ and use $y^k = x^k - d_k$ from (16):

$$\left\|x^{k+1} - x^\star\right\|^2 \leq \left\|x^k - x^\star\right\|^2 - 2\left\langle d_k,\, x^k - x^\star \right\rangle + \left\|d_k\right\|^2. \tag{21}$$

Cross term. Expanding d_k,

$$\left\langle d_k,\, x^k - x^\star \right\rangle = \mu_k\eta_k^{\mathrm{cvx}} \left\langle g_k^{\mathrm{cvx}},\, x^k - x^\star \right\rangle + (1-\mu_k)\eta_k^{\mathrm{ps}} R_k^{x^\star}. \tag{22}$$

If $\left\|g_k^{\mathrm{cvx}}\right\| = 0$ then $\eta_k^{\mathrm{cvx}} = 0$ and the first term vanishes. If $\left\|g_k^{\mathrm{cvx}}\right\| > 0$, Remark 3 gives $\left\langle g_k^{\mathrm{cvx}},\, x^k - x^\star \right\rangle \geq 2\Phi_k^{\mathrm{cvx}}$; multiplying by $\mu_k\eta_k^{\mathrm{cvx}} \geq 0$ and substituting $\eta_k^{\mathrm{cvx}} = 2\rho_{\mathrm{cvx}}\Phi_k^{\mathrm{cvx}} / \left\|g_k^{\mathrm{cvx}}\right\|^2$,

$$\mu_k\eta_k^{\mathrm{cvx}} \left\langle g_k^{\mathrm{cvx}},\, x^k - x^\star \right\rangle \geq \mu_k\rho_{\mathrm{cvx}} \frac{(2\Phi_k^{\mathrm{cvx}})^2}{\left\|g_k^{\mathrm{cvx}}\right\|^2} = \mu_k\rho_{\mathrm{cvx}} T_k. \tag{23}$$

Quadratic term. By Young's inequality $\|u+v\|^2 \leq (1+\epsilon)\left\|u\right\|^2 + (1+\epsilon^{-1})\left\|v\right\|^2$ with $u = \mu_k\eta_k^{\mathrm{cvx}} g_k^{\mathrm{cvx}}$, $v = (1-\mu_k)\eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}$,

$$\left\|d_k\right\|^2 \leq (1+\epsilon)\rho_{\mathrm{cvx}}^2\mu_k^2 T_k + (1+\epsilon^{-1})(1-\mu_k)^2(\eta_k^{\mathrm{ps}})^2 \left\|g_k^{\mathrm{ps}}\right\|^2.$$

By definition $\eta_k^{\mathrm{ps}} = \gamma_k / \max\{1, \left\|g_k^{\mathrm{ps}}\right\|\}$, so $\eta_k^{\mathrm{ps}} \left\|g_k^{\mathrm{ps}}\right\| \leq \gamma_k$ and hence $(\eta_k^{\mathrm{ps}})^2 \left\|g_k^{\mathrm{ps}}\right\|^2 \leq \gamma_k^2$; with $(1-\mu_k)^2 \leq 1$,

$$\left\|d_k\right\|^2 \leq (1+\epsilon)\rho_{\mathrm{cvx}}^2\mu_k^2 T_k + (1+\epsilon^{-1})\gamma_k^2. \tag{24}$$

Choose $\epsilon := (2-\rho_{\text{cvx}})/(2\rho_{\text{cvx}}) > 0$. Then $(1+\epsilon)\rho_{\text{cvx}} = (2+\rho_{\text{cvx}})/2$, so $(1+\epsilon)\rho_{\text{cvx}}^2 = \rho_{\text{cvx}}(2+\rho_{\text{cvx}})/2$, and $1+\epsilon^{-1} = (2+\rho_{\text{cvx}})/(2-\rho_{\text{cvx}}) = C_0$. Combining (21), (22), (23) and (24),

$$\left\|x^{k+1}-x^\star\right\|^2 \le \left\|x^k-x^\star\right\|^2 + \left[-2\mu_k\rho_{\text{cvx}} + (1+\epsilon)\rho_{\text{cvx}}^2\mu_k^2\right]T_k - 2(1-\mu_k)\eta_k^{\text{ps}}R_k^{x^\star} + C_0\gamma_k^2.$$

The bracket equals $\mu_k\rho_{\text{cvx}}\big(-2+\frac{2+\rho_{\text{cvx}}}{2}\mu_k\big)$. Since $\rho_{\text{cvx}} < 2$, for all $\mu_k \in [0,1]$ the factor $-2+\frac{2+\rho_{\text{cvx}}}{2}\mu_k$ is at most $-(2-\rho_{\text{cvx}})/2 < 0$ (its maximum over $[0,1]$ is at $\mu_k = 1$); multiplying by $\mu_k\rho_{\text{cvx}} \ge 0$ gives the bracket $\le -c_0\mu_k$. This yields (20). □

Lemma 3.9. *The sequence $\{x^k\}$ is bounded. Moreover, for each fixed $x^\star \in \Omega$:* (i) $\{\left\|x^k-x^\star\right\|^2\}$ *converges in* $[0,\infty)$*;* (ii) $\sum_{k=0}^\infty \mu_k T_k < \infty$*;* (iii) $\sum_{k=0}^\infty (1-\mu_k)\eta_k^{\text{ps}}R_k^{x^\star} < \infty$.

Proof. Fix $x^\star \in \Omega$ and set $a_k := \left\|x^k-x^\star\right\|^2$. By Lemma 3.5, $R_k^{x^\star} \ge 0$, and $\mu_k, T_k, (1-\mu_k)\eta_k^{\text{ps}} \ge 0$, so (20) gives $a_{k+1} \le a_k + C_0\gamma_k^2$. Since $\sum_k C_0\gamma_k^2 < \infty$ by (A3)(iii), Lemma 3.3 yields (i), and boundedness of $\{x^k\}$ follows. Summing (20) from 0 to $N-1$,

$$a_N + \sum_{k=0}^{N-1}\big(c_0\mu_k T_k + 2(1-\mu_k)\eta_k^{\text{ps}}R_k^{x^\star}\big) \le a_0 + C_0\sum_{k=0}^{N-1}\gamma_k^2.$$

The right side is bounded uniformly in N; letting $N \to \infty$ and using $c_0 > 0$ with nonnegative terms gives (ii) and (iii). □

Lemma 3.10. $\lim_{k\to\infty} J_k^{\text{cvx}} = 0$.

Proof. If $I^{\text{cvx}} = \emptyset$ then $J_k^{\text{cvx}} = 0$ for all k. Assume $I^{\text{cvx}} \ne \emptyset$ and put $B := \overline{\text{conv}}\{x^k : k \ge 0\}$, compact by Lemma 3.9 and contained in $K \subset U$. By (A2) and compactness,

$$L := \max_{i\in I^{\text{cvx}}}\sup_{x\in B}\|\nabla F_i(x)\| < \infty, \qquad V := \max_{i\in I}\sup_{x\in B}|F_i(x)-b_i| < \infty.$$

If $V = 0$ then all $v_i(x^k) = 0$ and $J_k^{\text{cvx}} = 0$; assume $V > 0$. If $L = 0$ then each F_i, $i \in I^{\text{cvx}}$, is constant on B, so $g_k^{\text{cvx}} = 0$ and, by Remark 3, $0 = \langle g_k^{\text{cvx}}, x^k - x^\star\rangle \ge 2\Phi_k^{\text{cvx}} \ge (J_k^{\text{cvx}})^2$, giving $J_k^{\text{cvx}} = 0$; assume $L > 0$. Set $m^{\text{cvx}} := |I^{\text{cvx}}| \ge 1$.

Consider a step with $J_k^{\text{cvx}} > 0$ and pick $i^\star \in I^{\text{cvx}}$ with $v_{i^\star}(x^k) = J_k^{\text{cvx}}$. Since $v_{i^\star}(x^k) \ge J_k^{\text{cvx}} - \delta$, we have $i^\star \in K_k^{\text{cvx}}$, whence

$$\Phi_k^{\text{cvx}} \ge \tfrac{1}{2}v_{i^\star}(x^k)^2 = \tfrac{1}{2}(J_k^{\text{cvx}})^2 > 0. \tag{25}$$

Fix $x^\star \in \Omega$. Remark 3 gives $\langle g_k^{\text{cvx}}, x^k - x^\star\rangle \ge 2\Phi_k^{\text{cvx}} > 0$, so $g_k^{\text{cvx}} \ne 0$; with Lemma 3.6, $\left\|g_k^{\text{cvx}}\right\|^2 \le 2m^{\text{cvx}}L^2\Phi_k^{\text{cvx}}$, and with (25),

$$T_k = \frac{(2\Phi_k^{\text{cvx}})^2}{\left\|g_k^{\text{cvx}}\right\|^2} \ge \frac{2\Phi_k^{\text{cvx}}}{m^{\text{cvx}}L^2} \ge \frac{(J_k^{\text{cvx}})^2}{m^{\text{cvx}}L^2}. \tag{26}$$

For a lower bound on μ_k: for all $i \in I$ and $x^k \in B$, $v_i(x^k) \le \left|F_i(x^k) - b_i\right| \le V$, so $J_k^{\mathrm{cvx}} \le V$ and $J_k^{\mathrm{ps}} \le V$; when $J_k^{\mathrm{cvx}} > 0$,

$$\mu_k = \frac{J_k^{\mathrm{cvx}}}{J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}}} \ge \frac{J_k^{\mathrm{cvx}}}{2V}. \tag{27}$$

Combining (26) and (27), for $J_k^{\mathrm{cvx}} > 0$, $\mu_k T_k \ge (J_k^{\mathrm{cvx}})^3/(2Vm^{\mathrm{cvx}}L^2)$; this also holds (as $0 \ge 0$) when $J_k^{\mathrm{cvx}} = 0$. By Lemma 3.9(ii),

$$\sum_{k=0}^{\infty} (J_k^{\mathrm{cvx}})^3 \le 2Vm^{\mathrm{cvx}}L^2 \sum_{k=0}^{\infty} \mu_k T_k < \infty, \tag{28}$$

so $(J_k^{\mathrm{cvx}})^3 \to 0$ and thus $J_k^{\mathrm{cvx}} \to 0$. □

Proposition 3.11. ~~$\sum_{k=0}^{\infty} \left\|x^{k+1} - x^k\right\|^2 < \infty;$~~

$$\sum_{k=0}^{\infty} \left\|x^{k+1} - x^k\right\|^2 < \infty;$$

in particular, $\left\|x^{k+1} - x^k\right\| \to 0$.

Proof. Since $x^k \in K$ and $x^{k+1} = P_K(y^k)$, Lemma 3.2 with $y = y^k$, $z = x^k$ gives $\left\|x^{k+1} - x^k\right\| \le \left\|y^k - x^k\right\| = \|d_k\|$. By (24) and $\mu_k^2 \le \mu_k$, $\|d_k\|^2 \le (1+\epsilon)\rho_{\mathrm{cvx}}^2 \mu_k T_k + C_0\gamma_k^2$. Summing and using Lemma 3.9(ii) and (A3)(iii), both series are finite. □

3.8. Strong convergence to a feasible point

The goal of the analysis is strong convergence of $\{x^k\}$ to a feasible point. The argument has three stages: exploit convex-track feasibility (Lemma 3.10) to confine the cluster set to the convex half; use a contradiction argument with pseudoconvexity to produce a cluster point in Ω; and combine with distance convergence to upgrade to convergence of the whole sequence.

Theorem 3.12. *Under Assumption 1, the sequence $\{x^k\}$ generated by Algorithm 1 converges strongly to a point $x^\infty \in \Omega$.*

Proof. By Lemma 3.9, $\{x^k\}$ is bounded; let $C := \{\bar{x} \in H : \exists\, k_l \to \infty,\ x^{k_l} \to \bar{x}\}$ be its (nonempty, compact) cluster set. Since K is closed and $x^k \in K$, $C \subseteq K$.

Stage 1 (convex half). Fix $\bar{x} \in C$ and $x^{k_l} \to \bar{x}$. By (A2) each F_i is continuous, so $v_i(x^{k_l}) \to v_i(\bar{x})$; for $i \in I^{\mathrm{cvx}}$, $v_i(x^{k_l}) \le J_{k_l}^{\mathrm{cvx}} \to 0$ by Lemma 3.10, so by squeezing $v_i(\bar{x}) = 0$, i.e. $F_i(\bar{x}) \le b_i$. Hence

$$C \subseteq \bigcap_{i \in I^{\mathrm{cvx}}} \{x \in H : F_i(x) \le b_i\}.$$

$$C \subseteq \bigcap_{i \in I^{\mathrm{cvx}}} \{x \in K : F_i(x) \le b_i\}. \tag{29}$$

If $I^{\mathrm{ps}} = \emptyset$, then (29) gives ~~$C \subseteq K \cap \bigcap_{i \in I^{\mathrm{cvx}}} \{F_i \le b_i\} = \Omega$~~$C \subseteq \Omega$, so $C \cap \Omega = C \neq \emptyset$. Assume henceforth $I^{\mathrm{ps}} \neq \emptyset$.

Stage 2 (pseudoconvex half by contradiction). Suppose, for contradiction, that $C \cap \Omega = \emptyset$. By (29) and the partition $I = I^{\mathrm{cvx}} \cup I^{\mathrm{ps}}$, for each $\bar{x} \in C$ there is $j \in I^{\mathrm{ps}}$ with $F_j(\bar{x}) > b_j$, i.e. $J^{\mathrm{ps}}(\bar{x}) > 0$. As a maximum of finitely many continuous functions, J^{ps} is continuous and positive on the compact set C, so $\alpha := \min_{x \in C} J^{\mathrm{ps}}(x) > 0$.

Claim 1 (uniform lower bound for J_k^{ps}). There exist $\tau_0 > 0$ and k_0 with $J_k^{\mathrm{ps}} \ge \tau_0$ for all $k \ge k_0$. Indeed, put $\tau_0 := \alpha/2$. If the claim failed there would be $k_l \to \infty$ with $J_{k_l}^{\mathrm{ps}} < \tau_0$; extracting (same notation) $x^{k_l} \to \tilde{x} \in C$, continuity gives $J^{\mathrm{ps}}(\tilde{x}) \le \tau_0 < \alpha$, contradicting the definition of α.

Claim 2 ($\mu_k \to 0$). For $k \ge k_0$, $\mu_k = J_k^{\mathrm{cvx}}/(J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}}) \le J_k^{\mathrm{cvx}}/\tau_0 \to 0$ by Lemma 3.10. Hence there is $k_1 \ge k_0$ with $1 - \mu_k \ge 1/2$ for $k \ge k_1$.

Claim 3 (uniform lower bound for $R_k^{x^\star}$). Fix $x^\star \in \Omega$. There exist $\xi_0 > 0$ and $k_2 \ge k_1$ with $R_k^{x^\star} \ge \xi_0$ for all $k \ge k_2$. Suppose not: there is $k_l \to \infty$ with $R_{k_l}^{x^\star} \to 0$. As $\{x^{k_l}\}$ is bounded, extract $x^{k_l} \to \bar{x} \in C$; since I^{ps} has finitely many subsets, extract further so that $K_{k_l}^{\mathrm{ps}} \equiv K^\star$ is constant. By assumption $J^{\mathrm{ps}}(\bar{x}) > 0$; choose $j^\star \in \arg\max_{j \in I^{\mathrm{ps}}} v_j(\bar{x})$, so $v_{j^\star}(\bar{x}) = J^{\mathrm{ps}}(\bar{x}) > 0$. By continuity $v_{j^\star}(x^{k_l}) \to v_{j^\star}(\bar{x})$ and $J_{k_l}^{\mathrm{ps}} \to J^{\mathrm{ps}}(\bar{x})$, so $J_{k_l}^{\mathrm{ps}} - v_{j^\star}(x^{k_l}) \to 0$; with $\delta > 0$ from (A3)(i), for large l we get $v_{j^\star}(x^{k_l}) \ge J_{k_l}^{\mathrm{ps}} - \delta$, hence $j^\star \in K_{k_l}^{\mathrm{ps}} = K^\star$. Each summand of $R_{k_l}^{x^\star} = \sum_{j \in K^\star} v_j(x^{k_l}) \langle \nabla F_j(x^{k_l}),\, x^{k_l} - x^\star \rangle$ is nonnegative by Lemma 3.5, so keeping only $j^\star$,

$$R_{k_l}^{x^\star} \ge v_{j^\star}(x^{k_l}) \left\langle \nabla F_{j^\star}(x^{k_l}),\, x^{k_l} - x^\star \right\rangle.$$

Letting $l \to \infty$ and using continuity of $v_{j^\star}$ and $\nabla F_{j^\star}$,

$$\liminf_{l \to \infty} R_{k_l}^{x^\star} \ge v_{j^\star}(\bar{x}) \langle \nabla F_{j^\star}(\bar{x}),\, \bar{x} - x^\star \rangle =: \Lambda.$$

Since $j^\star \in I^{\mathrm{ps}}$ and $x^\star \in \Omega$, $F_{j^\star}(x^\star) \le b_{j^\star} < F_{j^\star}(\bar{x})$; the contrapositive (5) with $x = \bar{x}$, $y = x^\star$ gives $\langle \nabla F_{j^\star}(\bar{x}),\, \bar{x} - x^\star \rangle > 0$, and with $v_{j^\star}(\bar{x}) > 0$ we get $\Lambda > 0$, contradicting $R_{k_l}^{x^\star} \to 0$. This proves Claim 3.

Reaching the contradiction. By Lemma 3.7 on $B := \overline{\mathrm{conv}}\{x^k\}$, $\|g_k^{\mathrm{ps}}\| \le |I^{\mathrm{ps}}|\, V_B L'$ with $L' := \max_{j \in I^{\mathrm{ps}}} \sup_B \|\nabla F_j\| < \infty$. Put $G' := \max\{1, |I^{\mathrm{ps}}|\, V_B L'\}$; then $\max\{1, \|g_k^{\mathrm{ps}}\|\} \le G'$ for all k, so $\eta_k^{\mathrm{ps}} = \gamma_k / \max\{1, \|g_k^{\mathrm{ps}}\|\} \ge \gamma_k / G'$. For $k \ge k_2$, combining with $1 - \mu_k \ge 1/2$ and $R_k^{x^\star} \ge \xi_0$,

$$\sum_{k=0}^{\infty} (1 - \mu_k) \eta_k^{\mathrm{ps}} R_k^{x^\star} \ge \sum_{k \ge k_2} \frac{1}{2} \cdot \frac{\gamma_k}{G'} \cdot \xi_0 = \frac{\xi_0}{2G'} \sum_{k \ge k_2} \gamma_k = +\infty$$

by (A3)(iii), contradicting Lemma 3.9(iii). Hence $C \cap \Omega \neq \emptyset$.

Stage 3 (whole-sequence convergence). Fix $\bar{x} \in C \cap \Omega$. ~~By~~ Lemma 3.9(i)~~with~~, applied to $x^\star = \bar{x}$, ~~$\|x^k - \bar{x}\|^2 \to \ell \ge 0$~~ ensures that $\lim_{k \to \infty} \|x^k - \bar{x}\|^2$ exists. Since $\bar{x} \in C$, ~~some subsequence $x^{k_l} \to \bar{x}$, so $\|x^{k_l} - \bar{x}\|^2 \to 0$, forcing $\ell = 0$; thus $x^k \to \bar{x}$. Set~~ a subsequence

x^{k_l} converges to $\bar{x}$. Therefore

$$\lim_{k\to\infty} \left\|x^k - \bar{x}\right\|^2 = \lim_{l\to\infty} \left\|x^{k_l} - \bar{x}\right\|^2 = 0.$$

Hence the whole sequence converges to $\bar{x}$, and we set $x^\infty := \bar{x} \in \Omega$. This argument requires neither uniqueness of the feasible point nor convexity of the target set: one feasible cluster point and convergence of its distance suffice. □

Remark 4. The parameter $\delta > 0$ is not necessary for convergence. When $\delta = 0$ one argues by subsequence: along $x^{k_l} \to \bar{x} \in C$, choose $j_l \in \arg\max_{j\in I^{\mathrm{ps}}} v_j(x^{k_l})$; since I^{ps} is finite, extract $j_l \equiv j^\star$, and $v_{j^\star}(x^{k_l}) = J^{\mathrm{ps}}(x^{k_l}) \to J^{\mathrm{ps}}(\bar{x}) > 0$, so $j^\star$ attains the maximal violation at $\bar{x}$ and the pseudoconvexity argument goes through. The role of $\delta > 0$ is to simplify: a maximizer at $\bar{x}$ automatically lies in $K^{\mathrm{ps}}_{k_l}$ for large l, avoiding a 'finitely many indices' extraction. Larger δ enlarges $K^{\mathrm{cvx}}_k, K^{\mathrm{ps}}_k$, which ~~aids~~ may aid numerical stability when several $v_i(x^k)$ are nearly equal; on the test problem (47), $\delta = 10^{-1}$ and $\delta = 10^{-3}$ produce identical iterates (Table 8): the pseudoconvex track has a single index, at every start either only F_1 is violated or $v_1 - v_2 \geq 4.79$, and in these runs the convex track is satisfied from the second evaluated point on. The δ row of Table 8 is therefore a consistency check, not evidence about δ.

Remark 5. From (28), $\sum_k (J^{\mathrm{cvx}}_k)^3 < \infty$, whence ~~$\min_{k\le N} J^{\mathrm{cvx}}_k = O(N^{-1/3})$~~$\min_{0\le k<N} J^{\mathrm{cvx}}_k = O(N^{-1/3})$. This bounds the *best* ~~convex~~ convex-track violation over the first N ~~steps, not J^{cvx}_N itself;~~ iterates. It is not a last-iterate bound, a rate for the full residual $J(x^k)$, or a rate for the distance to a feasible point. For these latter quantities, under (A1)–(A3) alone we assert only qualitative convergence.

3.9. Degenerate constraint structures

When the constraint structure degenerates (only a convex track or only a pseudoconvex track), DTAP and its analysis simplify, yielding two corollaries of Theorem 3.12.

Corollary 3.13. *If $I^{\mathrm{ps}} = \emptyset$, then $J^{\mathrm{ps}}_k \equiv 0$, $g^{\mathrm{ps}}_k \equiv 0$, and $\mu_k \equiv 1$ whenever $J^{\mathrm{cvx}}_k > 0$, so Algorithm 1 reduces to $x^{k+1} = P_K(x^k - \eta^{\mathrm{cvx}}_k g^{\mathrm{cvx}}_k)$. The sequence converges strongly to a point ~~$x^\infty \in \Omega = K \cap \bigcap_{i\in I^{\mathrm{cvx}}} \{F_i \le b_i\}$~~ $x^\infty \in \Omega = \{x \in K : F_i(x) \le b_i \text{ for all } i \in I^{\mathrm{cvx}}\}$.*

Proof. Here $d_k = \eta^{\mathrm{cvx}}_k g^{\mathrm{cvx}}_k$, ~~so no Young splitting is needed~~ and (21) ~~reads~~ directly gives $\left\|x^{k+1} - x^\star\right\|^2 \le \left\|x^k - x^\star\right\|^2 - 2\eta^{\mathrm{cvx}}_k \left\langle g^{\mathrm{cvx}}_k,\, x^k - x^\star\right\rangle + (\eta^{\mathrm{cvx}}_k)^2 \left\|g^{\mathrm{cvx}}_k\right\|^2$. If $\left\|g^{\mathrm{cvx}}_k\right\| = 0$ then $\eta^{\mathrm{cvx}}_k = 0$, $T_k = 0$, and the bound is $\left\|x^{k+1} - x^\star\right\|^2 \le \left\|x^k - x^\star\right\|^2$. Otherwise Lemma 3.4 gives $\left\langle g^{\mathrm{cvx}}_k,\, x^k - x^\star\right\rangle \ge 2\Phi^{\mathrm{cvx}}_k$, and substituting $\eta^{\mathrm{cvx}}_k = 2\rho_{\mathrm{cvx}}\Phi^{\mathrm{cvx}}_k / \left\|g^{\mathrm{cvx}}_k\right\|^2$ together with $(\eta^{\mathrm{cvx}}_k)^2 \left\|g^{\mathrm{cvx}}_k\right\|^2 = \eta^{\mathrm{cvx}}_k \cdot 2\rho_{\mathrm{cvx}}\Phi^{\mathrm{cvx}}_k$ yields, after simplification, $2\eta^{\mathrm{cvx}}_k \left\langle g^{\mathrm{cvx}}_k,\, x^k - x^\star\right\rangle - (\eta^{\mathrm{cvx}}_k)^2 \left\|g^{\mathrm{cvx}}_k\right\|^2 \ge \rho_{\mathrm{cvx}}(2 - \rho_{\mathrm{cvx}})(2\Phi^{\mathrm{cvx}}_k)^2 / \left\|g^{\mathrm{cvx}}_k\right\|^2 = 2c_0 T_k$. Hence

$$\left\|x^{k+1} - x^\star\right\|^2 \le \left\|x^k - x^\star\right\|^2 - 2c_0 T_k, \tag{30}$$

valid for all k. Thus $\{\left\|x^k - x^\star\right\|^2\}$ is nonincreasing and convergent, and summing gives

$\sum_k T_k < \infty$. As in Lemma 3.10, when $J_k^{\mathrm{cvx}} > 0$ one has $T_k \geq (J_k^{\mathrm{cvx}})^2/(m^{\mathrm{cvx}}L^2)$ (note the exponent 2 here, versus 3 in (28), since no μ_k factor is present), so $\sum_k (J_k^{\mathrm{cvx}})^2 < \infty$ and $J_k^{\mathrm{cvx}} \to 0$. Finally, $I^{\mathrm{ps}} = \emptyset$ makes Stage 2 of Theorem 3.12 vacuous (indeed $C \subseteq \Omega$ already from (29)), and Stage 3 gives $x^k \to x^\infty \in \Omega$. □

Corollary 3.14. *If $I^{\mathrm{cvx}} = \emptyset$, then $J_k^{\mathrm{cvx}} \equiv 0$, $g_k^{\mathrm{cvx}} \equiv 0$, and $\mu_k \equiv 0$ whenever $J_k^{\mathrm{ps}} > 0$, so Algorithm 1 reduces to $x^{k+1} = P_K(x^k - \eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}})$ with $\eta_k^{\mathrm{ps}} = \gamma_k / \max\{1, \|g_k^{\mathrm{ps}}\|\}$. The sequence converges strongly to a point $x^\infty \in \Omega$.*

Proof. Now $d_k = \eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}$ with $\|d_k\|^2 \leq \gamma_k^2$, so (21) gives $\|x^{k+1} - x^\star\|^2 \leq \|x^k - x^\star\|^2 - 2\eta_k^{\mathrm{ps}} R_k^{x^\star} + \gamma_k^2$, which is (20) in this degenerate case ($T_k = 0$, $1 - \mu_k = 1$). Since $R_k^{x^\star} \geq 0$ (Lemma 3.5), $a_{k+1} \leq a_k + \gamma_k^2$, and Lemma 3.3 gives convergence of $\{\|x^k - x^\star\|^2\}$ and boundedness; summation gives $\sum_k \eta_k^{\mathrm{ps}} R_k^{x^\star} < \infty$. Stage 1 of Theorem 3.12 gives $C \subseteq K$ (the convex part being vacuous); Stage 2 with $\mu_k = 0$ (so $1 - \mu_k = 1$) produces the same contradiction $\sum \eta_k^{\mathrm{ps}} R_k^{x^\star} = +\infty$, so $C \cap \Omega \neq \emptyset$; Stage 3 gives $x^k \to x^\infty \in \Omega$. □

~~[Former Section 5] A unified feature-selection framework~~

~~We apply the normal-box NSFP to feature selection. We build on two supervised models that have already been tested in practice. The first is the fractional-programming model of Wang et al. [32], which is solved by a projection neural network. The second is the fuzzy-rough-set model AFRSFCV of Wang et al. [33], which is solved by unconstrained softmax optimization. We add *no* new constraint of our own. Instead, we take the three constraint functions directly from the two models' objectives, namely two convex terms from the fuzzy-rough model and one pseudoconvex ratio from the fractional model. Together they form a single normal-box NSFP that satisfies Assumption 1, so DTAP can solve it. Throughout, $U = \{x_1, \ldots, x_n\}$ is the sample set, $A = \{a_1, \ldots, a_p\}$ the features, $X \in \mathbb{R}^{n \times p}$ the data matrix (column k being feature a_k), $y \in \{1, \ldots, r\}^n$ the class labels, and the weight vector w lives on the standard simplex $K := \Delta_{p-1} = \{w \in \mathbb{R}^p : w_k \geq 0,\ \mathbf{1}_p^\top w = 1\}$, a compact convex set that serves as the base set. Sections 4.1.1 and 4.1.2 recall the two source models and the quantities they produce. Section 4.1.3 then shows that these quantities give the three constraints of a single normal-box NSFP that DTAP can solve. Section 4.1.3 fixes the selected subset from the resulting weights. Finally, Section 5.5 adds a sparse-projection variant that enforces an explicit feature budget.~~

~~[Former Section 5.1] The fractional-programming model~~

~~[Former Section 5.1.1] Input quantities~~

[This paragraph of the submitted version (former Section 5.1.1) was moved; see p. 28]

~~[Former Section 5.1.2] The N-Fscore baseline~~

~~Following [32], supervised selection is the fractional program~~

$$\min_{w \in K} \frac{w^\top Q w}{\rho^\top w}, \qquad K = \Delta_{p-1},$$

~~Its numerator measures redundancy and its denominator relevance, so making the ratio small lowers redundancy while keeping relevance high. The ratio is pseudoconvex on K whenever $\rho_k > 0$ for all k [32]. Wang et al. solve it with a projection neural network (PNN) coupled to a k-winners-take-all stage. Our baseline *N-Fscore* runs only the PNN~~

stage with ρ set to the Fisher score of (33) and passes the resulting weights to the shared selection rule of Section 4.1.3, so it differs from the other methods only in how $w^\star$ is computed. The PNN dynamics and their explicit-Euler integration are given in Appendix A.

[Former Section 5.2] The fuzzy-rough-set model

Rough-set feature selection uses the uncertainty in the data to score features. In the survey of Liu et al. [24], these methods are organized along a model axis (classical, neighbourhood, fuzzy) and a measure axis (approximation, information, granularity). Fuzzy rough sets work on real-valued data directly, without discretization, and allow membership in $[0,1]$, which suits numeric data [20]. The AFRSFCV model of Wang et al. [33] belongs to this family and adds class variance to the fuzzy similarity relation.

[Former Section 5.2.1] Input quantities

Let $D=\{d\}$ be the decision attribute with r classes $D_1,\dots,D_r$, where $D_t:=\{x_i\in U : d(x_i)=t\}$. We write $a_k(x_i)$ for the value of x_i on a_k and $t(x_i)$ for its class index.

Definition 5.4 (Class-variance fuzzy similarity). With $\mu_{kt}:=\frac{1}{|D_t|}\sum_{x_i\in D_t}a_k(x_i)$ and $\sigma_{kt}^2:=\frac{1}{|D_t|}\sum_{x_i\in D_t}(a_k(x_i)-\mu_{kt})^2$, the *class-variance fuzzy similarity* of x_i,x_j on a_k is

$$R_k(x_i,x_j):=\exp\!\left(-\frac{[a_k(x_i)-a_k(x_j)]^2}{2\,\sigma_{k,t(x_i)}^2}\right).$$

Definition 5.5 (Fuzzy lower approximation). With the class indicator $D_t(x_j):=1$ if $x_j\in D_t$ and 0 otherwise, the *fuzzy lower approximation* of D_t via a_k is

$$\underline{R_k}(D_t)(x_i):=\min_{x_j\in U}\max\{1-R_k(x_i,x_j),\,D_t(x_j)\},\qquad x_i\in U,$$

collected into $\underline{R_k}(D)\in\mathbb{R}^{n\times r}$ with (i,t) entry $\underline{R_k}(D_t)(x_i)\in[0,1]$.

This measures the certainty of $x_i\in D_t$ using only a_k. Let $Y\in\{0,1\}^{n\times r}$ be the one-hot label matrix, $Y_{it}=1$ iff $x_i\in D_t$.

Definition 5.6 (Decision matrix). The *decision matrix* $P\in\{0,1\}^{n\times p}$ has

$$P_{ik}:=\begin{cases}1 & \text{if } \arg\max_{1\le t\le r}\underline{R_k}(D_t)(x_i)=t(x_i),\\ 0 & \text{otherwise,}\end{cases}$$

so $P_{ik}=1$ iff feature a_k classifies x_i correctly under the maximal lower-approximation rule.

[Former Section 5.2.2] The AFRSFCV baseline

[This paragraph of the submitted version (former Section 5.2.2) was moved; see p. 29]

[Former Section 5.3] Reduction to a nonlinear split feasibility problem

Both models seek a weight vector $w\in\mathbb{R}^p$ on the same feature set, with w_k encoding the relative importance of a_k. The normalization $w_k\ge 0$, $\sum_k w_k=1$ places w on the standard simplex $K=\Delta_{p-1}$, a compact convex set that serves as the base set of the split feasibility problem. From the two models' objectives we extract three constraint functions.

Definition 5.7 (The three constraint functions). Define

$$F_1(w) := \left\| Y - \sum_{k=1}^{p} w_k \underline{R_k}(D) \right\|_F^2,$$

$$F_2(w) := \left\| \mathbf{1}_n - Pw \right\|^2,$$

$$F_3(w) := \frac{w^\top Q w}{\rho^\top w},$$

where $Y, \underline{R_k}(D), P$ are from Section 4.1.2, Q, ρ from Section 4.1.1, and F_3 is defined on $\{w : \rho^\top w > 0\}$.

Here F_1, F_2 are the first two terms of the AFRSFCV loss (38), validated in [33], and F_3 is exactly the fractional objective (34) of [32]; no artificial function is added.

Proposition 5.8 (Constraint regularity). Under $Q \succeq 0$ and $\rho_k > 0$ for all k: F_1 and F_2 are convex on $\mathbb{R}^p$, and F_3 is pseudoconvex on K. Consequently, with $I^{\mathrm{cvx}} = \{1, 2\}$ and $I^{\mathrm{ps}} = \{3\}$, the feasibility problem

$$\text{find } w^\star \in \Omega, \qquad \Omega := \{w \in K : F_i(w) \le b_i,\ i = 1, 2, 3\},$$

with thresholds $b_1, b_2, b_3 > 0$, is an instance of the normal box NSFP (6) satisfying Assumption 1 (provided $\Omega \neq \emptyset$).

Proof. Each of F_1, F_2 is the squared norm of an affine map of w: indeed $w \mapsto Y - \sum_k w_k \underline{R_k}(D)$ and $w \mapsto \mathbf{1}_n - Pw$ are affine, and the squared norm of an affine map is convex; hence F_1, F_2 are convex on $\mathbb{R}^p$, meeting the convex-track requirement of (A1)(b). For F_3, the numerator $w^\top Q w$ is a positive semidefinite quadratic ($Q \succeq 0$ by (32)) and the denominator $\rho^\top w$ is a positive linear form on K; by [32] such a ratio is pseudoconvex on K, meeting the pseudoconvex-track requirement. All F_i are smooth with continuous gradients, giving (A2); (A1)(a) is the standing nonemptiness of Ω, and (A3) concerns only the algorithm parameters. Thus (42) is of the form (6).

By Proposition 4.1, DTAP (Algorithm 1) applies directly to (42). Although F_3 is only defined where $\rho^\top w > 0$, when $\rho_k > 0$ for all k we have $\rho^\top w = \sum_k \rho_k w_k \ge \min_k \rho_k > 0$ for every $w \in K$, so K lies in the open half-space and, since every DTAP iterate stays in K, both $F_3(w)$ and $\nabla F_3(w)$ are well defined and continuous along the sequence. As the convergence analysis in Section 3 uses constraint values only on K, Theorem 3.12 applies: the DTAP iterates for (42) converge strongly to a feasible $w^\star \in \Omega$.

Remark 4 (Per-iteration cost of the instance). For the instance (42) with n samples, p features, and r classes, the per-iteration cost of Section 3.5 is dominated by the constraint evaluations: $F_1, \nabla F_1$ from (39) cost $O(nrp)$ (a weighted combination of the p matrices $\underline{R_k}(D)$); $F_2, \nabla F_2$ from (40) cost $O(np)$ through P; $F_3, \nabla F_3$ from (41) cost $O(p^2)$ through the quadratic form $w^\top Q w$; and the simplex projection costs $O(p \log p)$. One DTAP iteration thus runs in $O(nrp + p^2)$ time and $O(nrp)$ memory (the lower-approximation matrices), the same per-iteration order as the AFRSFCV and Hu subgradient baselines, which operate on the same matrices; the difference lies in the *number* of iterations, quantified in Section 6.4 (Table B3).

Problem (42) can also be solved by the subgradient method for *quasi-convex* feasibility of Hu et al. [19]: since convex and pseudoconvex functions are quasi-convex, all three constraints are quasi-convex on K, so (42) is a quasi-convex feasibility problem. We use two variants as baselines, *SG-const* (constant step) and *SG-dyn* (dynamic step), both

following the most-violated-constraint control scheme (at tolerance level α) to form a search direction, then updating and projecting onto K. Unlike DTAP, the SG-dyn variant must estimate a Hölder modulus of the constraint functions per dataset, a cost that DTAP avoids through the Polyak step on the convex track.

[Former Section 5.4] From weights to a selected subset

[This paragraph of the submitted version (former Section 5.4) was moved; see p. 31]

[Former Section 5.5] Explicit feature budgets via sparse projection

The cumulative-threshold rule (44) selects features by how *concentrated* $w^\star$ is, not by their number. On high-dimensional data this is a limitation: because the convex constraints F_1, F_2 spread weight across many features to reproduce the label structure, $w^\star$ is nearly flat, so reaching cumulative mass θ forces most coordinates to be kept. At $\theta = 0.9$, DTAP retains 33/36 features on Satimage and 49/57 on Spambase (Table 2), contrary to the goal $|A^\star| \ll p$. We remove this limitation by moving the sparsity requirement *inside* the iteration.

Definition 5.9 (s-sparse simplex). For a budget $s \in \{1, \dots, p\}$, the s-sparse simplex is $\Delta_{p-1}^{(s)} := \{w \in \Delta_{p-1} : \|w\|_0 \le s\}$, where $\|w\|_0 = |\{k : w_k \neq 0\}|$.

Although $\Delta_{p-1}^{(s)}$ is nonconvex, a union of the s-dimensional faces of the simplex, its Euclidean projection has a closed form via the Greedy Selector and Simplex Projector (GSSP) [22]: for $y \in \mathbb{R}^p$, let $S(y)$ index the s largest entries of y; then

$$\left(P_{\Delta_{p-1}^{(s)}}(y)\right)_k = \begin{cases} \left(P_{\Delta_{s-1}}(y_{S(y)})\right)_k & k \in S(y), \\ 0 & \text{otherwise}, \end{cases}$$

that is, keep the s largest coordinates, project them onto the standard simplex, and zero the rest; the cost is $O(p \log p)$, the same order as P_K.

The DTAP-S algorithm. Replacing the base projection in Step 2 of Algorithm 1 with $w^{k+1} = P_{\Delta_{p-1}^{(s)}}(y^k)$ yields the sparse variant DTAP-S$_s$. Every iterate then carries at most s nonzeros, so the selected set is exactly $\mathrm{supp}(w^\star)$ with $|A^\star| \le s$: the budget s is an *explicit, tunable* feature count that replaces the indirect threshold θ.

Remark 5 (A deliberate heuristic trade-off). Since $\Delta_{p-1}^{(s)}$ is nonconvex, the projection (46) is no longer nonexpansive, so the strong-convergence guarantee of Theorem 3.12 does not transfer; DTAP-S is thus a heuristic that trades the convex-case guarantee for genuine feature reduction. Empirically (Section 6.3) the iterates remain stable and the best iterate is retained. The budget s caps the support while the dual-track update is otherwise unchanged, so all the modelling of Section 4.1.3 is preserved.

In summary, the redundancy and Fisher quantities of the fractional model and the lower-approximation, decision, and label matrices of the fuzzy-rough model jointly furnish the three constraints of a single nonlinear split feasibility problem. The reduction reuses only validated objectives, preserving the meaning of both source models while bringing the problem into a form solvable by DTAP (either exactly, via the strongly convergent base algorithm, or under an explicit budget, via the sparse variant DTAP-S) and remaining open to new selection criteria added as further convex or pseudoconvex constraints.

4. Numerical experiments

We separate predictive performance from the numerical solution of the feasibility system. The classification comparison uses eight datasets. The solver diagnostic additionally includes Movement Libras and controlled nonconvex pseudoconvex instances. These two scopes are stated separately below.

4.1. Feature-selection model

We apply the normal-box NSFP to feature selection by combining criteria from the fractional-programming model of Wang et al. [32] and the fuzzy-rough model AFRS-FCV [33]. Let $\mathcal{X} = \{x_1, \dots, x_n\}$ be the training samples, $A = \{a_1, \dots, a_p\}$ the features, $X \in \mathbb{R}^{n \times p}$ the data matrix, and $y \in \{1, \dots, r\}^n$ the class labels. The feature weights belong to the compact convex simplex $K = \Delta_{p-1} = \{w \in \mathbb{R}^p : w \geq 0,\ \mathbf{1}_p^\top w = 1\}$. All supervised quantities below are estimated from the training data.

4.1.1. Fractional criterion and N-Fscore

Definition 5.1 (Supervised similarity matrix). The *supervised similarity matrix* $S \in \mathbb{R}^{p \times p}$ has entries

$$s_{ij} = \max\left\{0,\ \frac{I(F_i; F_j; y)}{H(F_i) + H(F_j)}\right\}, \qquad i, j = 1, \dots, p,$$

where $I(F_i; F_j; y) = I(F_i; y) + I(F_j; y) - I(F_i, F_j; y)$ is the multi-information, $H(\cdot)$ is entropy, and all quantities are estimated from equal-width histograms with K_{bin} bins per feature [4].

Definition 5.2 (Redundancy matrix). The *redundancy matrix* $Q \in \mathbb{R}^{p \times p}$ is obtained by a diagonal shift

$$Q := \delta_Q I_p + S, \qquad \delta_Q \geq -\min\{0, \lambda_{\min}(S)\},$$

which guarantees $Q \succeq 0$.

The condition $Q \succeq 0$ makes the quadratic form $w^\top Q w$ convex on $\mathbb{R}^p$.

Definition 5.3 (Fisher-score vector). The *Fisher-score vector* $\rho \in \mathbb{R}^p$ has components

$$\rho_k := \frac{\sum_{t=1}^{r} n_t (\mu_{kt} - \mu_k)^2}{\sum_{t=1}^{r} n_t\, \sigma_{kt}^2 + \varepsilon_\rho}, \qquad k = 1, \dots, p,$$

where n_t is the size of class t, μ_{kt}, σ_{kt}^2 are the class-t mean and variance of a_k, μ_k is the overall mean, and $\varepsilon_\rho > 0$ avoids division by zero.

The supervised similarity matrix $S \in \mathbb{R}^{p \times p}$ has entries

$$s_{ij} = \max\left\{0,\ \frac{I(Z_i; Z_j; C)}{H(Z_i) + H(Z_j)}\right\}, \qquad i, j = 1, \dots, p, \tag{31}$$

where Z_i is the discretized feature a_i, C is the class variable, and $I(Z_i; Z_j; C) = I(Z_i; C) + I(Z_j; C) - I((Z_i, Z_j); C)$. Entropy and mutual information are estimated

from equal-width histograms with ten bins per feature [4]. We set $s_{ij} = 0$ if the denominator is zero. The redundancy matrix is

$$Q_R = S + \delta_Q I_p, \qquad \delta_Q \geq -\min\{0, \lambda_{\min}(S)\}, \tag{32}$$

so $Q_R \succeq 0$. The implementation uses $\delta_Q = \max\{0, -\lambda_{\min}(S)\} + 10^{-10}$. Let n_t, $\bar{a}_{kt}$, and σ_{kt}^2 be the class size, mean, and population variance of feature a_k in class t, and let $\bar{a}_k$ be its overall mean. The Fisher scores are

$$\rho_k = \frac{\sum_{t=1}^{r} n_t(\bar{a}_{kt} - \bar{a}_k)^2}{\sum_{t=1}^{r} n_t \sigma_{kt}^2 + \varepsilon_\rho}, \qquad \varepsilon_\rho = 10^{-12}. \tag{33}$$

These scores are nonnegative. The denominator regularization prevents division by zero but does not ensure $\rho_k > 0$, since the numerator can vanish.

The fractional program underlying N-Fscore is

$$\min_{w \in K} \frac{w^\top Q_R w}{\rho^\top w}, \tag{34}$$

assuming the denominator is positive on K. The numerator measures redundancy and the denominator relevance. Our N-Fscore baseline uses the projection-neural-network stage of [32], with the Fisher scores in (33). It then applies the common selection rule (44), instead of the source method's k-winners-take-all stage. Its dynamics and explicit-Euler implementation are given in Section 4.2.1.

4.1.2. Fuzzy-rough criterion and AFRSFCV

For each class, write $D_t = \{x_i : y_i = t\}$ and $D_t(x_j) = \mathbf{1}_{\{y_j = t\}}$. AFRSFCV incorporates class variance into a fuzzy relation. With $a_k(x_i) = X_{ik}$, the implemented relation is

$$R_k(x_i, x_j) = \exp\left(-\frac{[a_k(x_i) - a_k(x_j)]^2}{2\widetilde{\sigma}_{k,y_i}^2}\right), \qquad \widetilde{\sigma}_{kt}^2 = \max\{\sigma_{kt}^2, 10^{-10}\}. \tag{35}$$

The variance floor makes the formula defined when a feature is constant within a class. The fuzzy lower approximation is

$$\underline{R_k}(D_t)(x_i) = \min_{x_j \in \mathcal{X}} \max\{1 - R_k(x_i, x_j), D_t(x_j)\}. \tag{36}$$

Its values form the matrix $\underline{R_k}(D) \in [0,1]^{n \times r}$. Let $Y_{it} = \mathbf{1}_{\{y_i = t\}}$ be the label matrix and define the decision matrix by

$$P_{ik} = \mathbf{1}_{\{\arg\max_{1 \leq t \leq r} \underline{R_k}(D_t)(x_i) = y_i\}}. \tag{37}$$

Ties are resolved by the smallest class index. Thus $P_{ik} = 1$ precisely when feature a_k predicts sample x_i correctly under this rule.

~~AFRSFCV [33] uses $\underline{R_k}(D)$, P, and Y, and minimizes, over a softmax parametriza-~~

~~tion,~~

$$\min_{\tau\in\mathbb{R}^p} L(\tau) := \frac{1}{2}\left\|Y - \sum_{k=1}^{p} w_k(\tau)\underline{R_k}(D)\right\|_F^2 + \frac{\lambda_1}{2}\left\|\mathbf{1}_n - Pw(\tau)\right\|^2 + \frac{\lambda_2}{2}\|w(\tau)\|^2,$$

where $\|\cdot\|_F$ is the Frobenius norm and the softmax $w_k(\tau) = e^{\tau_k}/\sum_{l=1}^{p} e^{\tau_l}$ keeps $w(\tau) \in K = \Delta_{p-1}$. The first term measures the gap between the label matrix Y and the lower-approximation combination. The second penalizes wrong-class confidence, where $\sum_k P_{ik} w_k$ is the confidence of a correct call. The third is Tikhonov regularization. Our baseline minimizes $L(\tau)$ by gradient descent for a fixed budget and passes the final weight $w^\star = w(\tau^T) \in K$ to the shared selection rule of Section 4.1.3. The gradient and update are given in Appendix A.

The AFRSFCV baseline [33] minimizes

$$\min_{\tau\in\mathbb{R}^p} L(\tau) = \frac{1}{2}\left\|Y - \sum_{k=1}^{p} w_k(\tau)\underline{R_k}(D)\right\|_F^2 + \frac{\lambda_1}{2}\|\mathbf{1}_n - Pw(\tau)\|^2 + \frac{\lambda_2}{2}\|w(\tau)\|^2, \tag{38}$$

where $w_k(\tau) = e^{\tau_k}/\sum_{l=1}^{p} e^{\tau_l}$ ensures $w(\tau) \in K$. The three terms measure label-approximation error, incorrect-decision confidence, and Tikhonov regularization, respectively. The baseline uses gradient descent for a fixed iteration budget, followed by (44). The gradient and update are given in Section 4.2.1.

4.1.3. Feasibility and feature selection

We impose separate bounds on three criteria,

$$F_1(w) = \left\|Y - \sum_{k=1}^{p} w_k \underline{R_k}(D)\right\|_F^2, \tag{39}$$

$$F_2(w) = \|\mathbf{1}_n - Pw\|^2, \tag{40}$$

$$F_3(w) = \frac{w^\top Q_R w}{\rho^\top w}, \qquad D_\rho = \{w : \rho^\top w > 0\}. \tag{41}$$

Here F_1, F_2 are the first two components of (38), without their coefficients, and F_3 is the objective in (34). For thresholds $b_i > 0$, the resulting problem is

$$\text{find } w^\star \in \Omega, \qquad \Omega = \{w \in K : F_i(w) \le b_i,\ i = 1, 2, 3\}. \tag{42}$$

This formulation asks for simultaneous acceptability of the three criteria. It omits the Tikhonov term and does not preserve either source optimization problem or its minimizers. A feasible weight need not minimize either baseline objective.

Proposition 4.1. *If $Q_R \succeq 0$ and $\min_k \rho_k > 0$, all three functions in (39)–(41) are smooth and convex on an open convex neighbourhood of K. Consequently, (42) is an all-convex normal-box NSFP. It satisfies the function assumptions of Assumption 1 whenever $\Omega \neq \varnothing$.*

Proof. The first two functions are squared norms of affine maps. For $w \in D_\rho$ and any direction h, put $t = \rho^\top w$ and $a = \rho^\top h$. Direct differentiation gives

$$D^2 F_3(w)[h,h] = \frac{2}{t}\left(h - \frac{a}{t}w\right)^\top Q_R \left(h - \frac{a}{t}w\right) \geq 0. \tag{43}$$

Thus F_3 is convex and smooth on the open half-space D_ρ. Since $\rho^\top w \geq \min_k \rho_k > 0$ for $w \in K$, this half-space contains K. □

The assignment $I^{\mathrm{cvx}} = \{1,2\}$, $I^{\mathrm{ps}} = \{3\}$ used in the experiments is admissible because convexity implies pseudoconvexity. It treats F_3 conservatively and does not test a genuinely nonconvex constraint. The all-convex assignment $I^{\mathrm{cvx}} = \{1,2,3\}$ is also available. Under the assumptions of Theorem 3.12, either assignment converges to a feasible weight. Score positivity must be checked on each instance. Replacing a zero score by a positive floor changes the model and must be declared. The implementation also uses $\max\{\rho^\top w, 10^{-15}\}$ during evaluation. This agrees with (41) only where the floor is inactive. Applying the theorem to this implementation therefore requires that the floor remain inactive.

Externally specified thresholds remain fixed. Training-based thresholds instead define the benchmark instance and must be shared by all solvers. Separate empirical percentiles do not ensure $\Omega \neq \varnothing$. A common point must satisfy all constraints, and checks to a numerical tolerance establish only the corresponding approximate feasibility. The threshold protocol and constraint-activity diagnostics are reported with the experiments.

~~Once a weight vector $w^\star$ is obtained by any method, the selected feature set is fixed by a shared *cumulative-threshold* rule. Sorting the weights in decreasing order $w^\star_{(1)} \geq \cdots \geq w^\star_{(p)}$, take the smallest count s whose cumulative mass reaches a threshold $\theta \in (0,1)$:~~

$$~~s := \min\Big\{ s' \in \{1,\dots,p\} : \sum_{j=1}^{s'} w^\star_{(j)} \geq \theta \Big\},~~$$

~~and keep the s features with the largest weights. The same rule is applied to every method, so that differences in results come only from the weight-computation stage.~~

For every method, sort the computed weights as $w^\star_{(1)} \geq \cdots \geq w^\star_{(p)}$ and retain the first s features, where

$$s = \min\left\{ s' \in \{1,\dots,p\} : \sum_{j=1}^{s'} w^\star_{(j)} \geq \theta \right\}, \qquad 0 < \theta < 1. \tag{44}$$

This cumulative-weight rule does not enforce a prescribed feature budget. Truncating the weights does not preserve the feasibility guarantee for the full vector.

Remark 6. With the precomputed matrices, evaluating $F_1, \nabla F_1$ costs $O(nrp)$, evaluating $F_2, \nabla F_2$ costs $O(np)$, and evaluating $F_3, \nabla F_3$ costs $O(p^2)$. Simplex projection costs $O(p \log p)$. A dense iteration therefore costs $O(nrp + p^2)$ operations and requires $O(nrp + p^2)$ storage, including Q. Preprocessing and classifier training are separate costs.

We evaluate DTAP on the feature-selection problem (42) against four baselines, namely the two base models N-Fscore [32] and AFRSFCV [33] and the two subgradient variants SG-const and SG-dyn of Hu et al. [19]. The experiments use eight UCI benchmarks that vary in sample size, dimension, class count, and balance. We report selected-set size, classification accuracy, area under the ROC curve, and selection stability, together with runtime and convergence behaviour.

4.2. Experimental setup

The main classification study uses the eight UCI datasets [21] in Table 1. The DTAP runs reported here start at the uniform weight vector and use $\rho_{\mathrm{cvx}} = 1.98$, $\delta = 10^{-3}$, $\gamma_k = (k+0.5)^{-0.55}$, tolerance 10^{-6}, and a budget of 1000 iterations. SG-const uses step 10^{-3}. The dynamic subgradient comparison is considered separately with sampled and empirically tuned moduli in the solver diagnostic. N-Fscore uses $\varepsilon_{\mathrm{PNN}} = 10^{-3}$ and 1000 Euler grid points on $[0, 0.2]$. AFRSFCV uses learning rate 10^{-3}, $\lambda_1 = 1$, $\lambda_2 = 0.5$, and 1000 steps. The shared cumulative threshold is $\theta = 0.9$.

[Former Section 6.1.1] Datasets

Eight UCI datasets [21] are used, with n from a few hundred to over ten thousand, p from 7 to 57, and r from 2 to 10 (Table 1).

Table 1. UCI datasets Datasets used for the main classification comparison. The additional Movement Libras instance is used only in the experimentssolver diagnostic.

Dataset	Samples n	Features p	Classes r	Balance
Glass	214	9	6	Imbalanced
Diabetes	768	8	2	Imbalanced
Segmentation	2 310	18	7	Balanced
Abalone	4 177	8	3	Balanced
Satimage	6 435	36	6	Imbalanced
Yeast	1 484	7	10	Imbalanced
Spambase	4 601	57	2	Imbalanced
Pendigits	10 992	16	10	Balanced

4.2.1. Baseline solvers

This appendix records the solver mechanics of the two base models, deferred from Sections 4.1.1 and 4.1.2 for readability.

The following updates specify the two baseline solvers for the models in Sections 4.1.1 and 4.1.2.

N-Fscore: the projection neural network To solve the fractional program (34), Wang et al. [32] proposed a neurodynamic method coupling a projection neural network (PNN) with a k-winners-take-all network. The PNN is the differential system

$$\varepsilon_{\mathrm{PNN}} \frac{du}{dt} = -Pg(u) - (I - P)\big(u - g(u) + \nabla f((I - P)g(u) + \tfrac{1}{p}\mathbf{1}_p)\big) + \tfrac{1}{p}\mathbf{1}_p,$$

$$\varepsilon_{\mathrm{PNN}}\frac{du}{dt} = -\Pi g(u) + \tfrac{1}{p}\mathbf{1}_p - (I-\Pi)\big(u - g(u) + \nabla f((I-\Pi)g(u) + \tfrac{1}{p}\mathbf{1}_p)\big). \tag{45}$$

with output $w = g(u)$, where $g(u) = \max\{0, u\}$ is the componentwise ReLU, ~~$P = \frac{1}{p}\mathbf{1}_p\mathbf{1}_p^\top\Pi = \frac{1}{p}\mathbf{1}_p\mathbf{1}_p^\top$~~, ∇f is the gradient of the objective in (34), and $\varepsilon_{\mathrm{PNN}} > 0$ is a time constant. We integrate (45) by explicit Euler on $[0, T]$ with N_{step} steps under a fixed budget (no stopping rule), with ~~T, N_{step} large enough for w to stabilize~~the fixed parameters reported in the experimental protocol, and take as weight vector the projection $P_K\big(g(u(T))\big)$ of the network output at the final time onto the simplex K. For a uniform comparison we use only the PNN part to obtain $w^\star$; the k-WTA stage is replaced by the shared cumulative-threshold rule of Section 4.1.3.

AFRSFCV: gradient and update The softmax loss (38) is minimized over $\tau \in \mathbb{R}^p$ by gradient descent. Its gradient is

$$\nabla_\tau L(\tau) = \big(\mathrm{diag}(w(\tau)) - w(\tau)w(\tau)^\top\big)\nabla_w L(w(\tau)), \tag{46}$$

with $\nabla_w L = -\mathbf{M}^\top(\mathrm{vec}(Y) - \mathbf{M}w) + \lambda_1(P^\top Pw - P^\top\mathbf{1}_n) + \lambda_2 w$, where $\mathbf{M} \in \mathbb{R}^{nr\times p}$ has k-th column $\mathrm{vec}(\underline{R_k}(D))$. The update $\tau^{t+1} = \tau^t - \eta\,\nabla_\tau L(\tau^t)$ runs for a fixed budget of T steps; the final weight $w^\star = w(\tau^T) \in K$ feeds the shared selection rule of Section 4.1.3.

~~[Former Section 6.1.2] Parameters and shared inputs~~

~~On each training split, features are min–max scaled to $[0, 1]$, and the inputs shared by all five methods are computed once. These are the fuzzy lower-approximation matrices $\underline{R_k}(D)$, the label matrix Y, and the decision matrix P (Section 4.1.2), together with the redundancy matrix Q and the Fisher vector ρ (Section 4.1.1); multi-information and entropy are estimated from equal-width histograms with 10 bins per feature [4]. The DTAP parameters are fixed across all datasets. We start from $w^0 = (1/p, \ldots, 1/p)^\top$ and use the Polyak parameter $\rho_{\mathrm{cvx}} = 1.98$, the pseudoconvex step $\gamma_k = 1/(k + 0.5)^{0.55}$ (so $\sum_k \gamma_k = \infty$ and $\sum_k \gamma_k^2 < \infty$), the approximate-active parameter $\delta = 10^{-3}$, the tolerance $\varepsilon = 10^{-6}$, and at most 1 000 iterations. The Hu variants use the same 1 000-iteration cap and the most-violated-constraint control scheme, where SG-const sets $v = 10^{-3}$ and SG-dyn estimates a per-dataset Hölder modulus by Dirichlet sampling on the simplex. N-Fscore uses $\varepsilon_{\mathrm{PNN}} = 10^{-3}$, $T = 0.2$, and $N_{\mathrm{step}} = 1\,000$ Euler steps under a fixed budget. AFRSFCV uses learning rate $\eta = 10^{-3}$, weights $\lambda_1 = 1$, $\lambda_2 = 0.5$, and $T = 1\,000$ gradient steps. The shared cumulative threshold is $\theta = 0.9$. The three DTAP parameters have transparent roles. The Polyak relaxation $\rho_{\mathrm{cvx}} \in (0, 2)$ trades per-step decrease against the perturbation constant (we use 1.98 for fast early progress), the exponent $s \in (1/2, 1]$ controls the safeguarded step ($s = 0.55$), and δ only stabilizes the search directions when several violations are nearly equal (Remark 4). These settings are structural rather than tuned per dataset, so one configuration serves the whole suite. Since all five methods consume the same precomputed inputs, any difference in the outputs arises only in the weight-computation stage and the shared threshold rule (44). Further implementation and reproducibility details are given in Appendix B.~~

~~[Former Section 6.1.3] Data-driven output thresholds~~

~~The thresholds b_1, b_2, b_3 in Ω (42) are set by a data-driven calibration. For each dataset we draw 3 000 points $w^{(s)} \sim \mathrm{Dir}(\mathbf{1}_p)$ on the simplex K and set b_i to a per-~~

~~dataset percentile of the sampled values $F_i(w^{(s)})$, tuned so that $\Omega \neq \emptyset$ yet is tight enough for the reduction to be effective. Since F_1, F_2 are sums of squared residuals over n samples, b_1, b_2 grow with sample size, whereas b_3, a ratio, stays small. The full procedure and the resulting thresholds are given in Appendix B (Table B1).~~

4.2.2. Evaluation protocol

Each dataset uses a single stratified 80/20 training-pool/test split (seed 42). Min–max scaling is fitted once on the full training pool, not separately within its selection folds, and is then applied to the test data. Stratified folds (seed 42) within the pool use $N_f = 10$, reduced to 7 on Glass and 4 on Yeast. The fold masks use $\theta = 0.9$ in (44); the fixed strict-majority rule retains features selected in at least $\lfloor N_f/2 \rfloor + 1$ folds. The final classifiers use the full pool with these voted features: RBF-SVM ($C = 1$, `gamma=scale`) and 5-NN, with the full feature set as a reference. Accuracy and macro one-vs-rest AUC are reported.

The voted masks are computed once and then reused: the evaluation stage trains only the downstream classifiers on these fixed masks, with SVM probability-calibration seed 42; identical masks share the same predictions. No classifier parameter or vote threshold is chosen from these test outcomes. The same holdouts were used when the selections were computed, and the threshold heuristic uses the whole-dataset sample size, dimensions and class counts before splitting. Hence this is an exploratory evaluation, not an end-to-end or nested-validation study. For the fold-selection matrix $Z \in \{0,1\}^{N_f \times p}$, the Nogueira index [28] is computed as

$$\Psi = 1 - \frac{p^{-1}\sum_{j=1}^{p} s_j^2}{\bar{q}(1-\bar{q})}, \quad s_j^2 = \frac{1}{N_f - 1}\sum_{b=1}^{N_f}(Z_{bj} - \bar{Z}_j)^2, \quad \bar{q} = \frac{1}{N_f p}\sum_{b,j} Z_{bj}.$$

Here $\bar{Z}_j = N_f^{-1}\sum_b Z_{bj}$; the unbiased variance correction is $N_f/(N_f - 1)$, not $p/(p-1)$. Since the training folds overlap, stability is reported descriptively. For each classifier metric, we use a two-sided signed-rank test across the eight datasets, omitting zero differences, assigning midranks to ties, and enumerating all signs of the nonzero differences; all-zero pairs have $p = 1$. Holm adjustment covers the four DTAP–baseline comparisons separately for each metric. The Friedman test compares the five selection methods.

~~Each dataset is split 80/20 in a stratified manner into a development pool and a held-out test set. On the pool we run stratified K-fold cross-validation with $K = 10$, reduced to the size of the minority class when needed ($K = 7$ on Glass, $K = 4$ on Yeast). In each fold, after scaling and input computation, the five methods are run independently to obtain weight vectors, and rule (44) with $\theta = 0.9$ fixes the selected set; a feature is kept if selected in more than half the folds (majority vote). Selection stability across folds is measured by the Nogueira index $\Phi \in [-1, 1]$ [28], with $\Phi = 1$ for perfectly stable selection. The voted feature set then trains two classifiers on the full pool, which are evaluated on the held-out test set. These are an RBF-kernel support vector machine (RBF-SVM, $C = 1$) [12] and a 5-nearest-neighbour classifier (5-NN) [13]. The two classification metrics are accuracy and macro one-vs-rest AUC. For reference, both classifiers are also evaluated on the full feature set.~~

~~Discrimination is measured by the macro one-vs-rest AUC, $\frac{1}{r}\sum_c \mathrm{AUC}_c$ [17], and a selection is conventionally called stable when $\Phi \gtrsim 0.75$. Across the eight datasets we compare methods by two-sided Wilcoxon signed-rank and Friedman tests [14] at~~

~~$\alpha = 0.05$, with Holm–Bonferroni correction for the family-wise error.~~

4.3. Classification results

Tables 2–4 report selected-set size, stability, and predictive performance. DTAP retains 84.2% of the features on average, including 33/36 on Satimage and 49/57 on Spambase. Its mean stability is 0.786, compared with N-Fscore's 0.822; thus the results support moderate reduction rather than uniform sparsity or stability superiority. AFRSFCV selects much smaller sets on several datasets, with lower mean accuracy under the fixed configuration.

DTAP's mean accuracy is within 0.44 percentage points of the full-feature reference for both classifiers. It exceeds N-Fscore and AFRSFCV by 3.12 and 4.83 points on SVM accuracy, while retaining more features. This is a size–prediction tradeoff, not a matched-budget comparison. Both SG variants have exactly the same voted masks as DTAP on seven datasets; only Spambase differs (48 versus 49 features). SG-dyn used the full 1000-iteration budget in 6/7 selection folds on Glass, 7/10 on Diabetes and 10/10 on Abalone, and SG-const in 1/10 on Abalone (DTAP in none); the voted weights of both SG variants are the best iterates found within the budget. The per-dataset exceptions are included in Table 3, and all five methods appear in the summary. Fold-wise stability can differ even when the final voted mask is identical.

Table 5 reports the signed-rank tests. None of the pairwise comparisons rejects at 0.05 after Holm adjustment; the SVM comparisons with the two base models have adjusted p-values between 0.0625 and 0.0938. The asymptotic Friedman tests reject equality globally, but this does not establish a particular pairwise superiority. All DTAP–SG pairwise p-values equal 1; this reflects ties and limited evidence, not a statistical equivalence claim. These small-sample comparisons concern eight heterogeneous datasets and do not measure variation under repeated train/test splits.

~~We analyse selection and classification along four axes, namely selected-set size, accuracy, AUC, and selection stability. As the statistical tests at the end of this subsection show, DTAP and the two Hu variants give almost identical results, with Wilcoxon $p = 1$ on three of the four metrics and only a small difference on RBF-SVM AUC; the tables below therefore list DTAP as the representative of the split-feasibility group, with SG-const and SG-dyn compared separately in Table 5 and in Section 6.4.~~

Former Table 3. Number of selected features $|A^\star|$ and selection stability (Nogueira index) across folds. Column p is the original feature count. The highest stability among the three selection methods is in bold.

Dataset	p	Selected $\lvert A^\star\rvert$			Nogueira stability		
		N-Fscore	AFRSFCV	DTAP	N-Fscore	AFRSFCV	DTAP
Glass	9	7	7	7	**0.8229**	0.6230	0.6458
Diabetes	8	7	3	7	0.2698	**0.8050**	0.6989
Segmentation	18	8	2	16	0.8904	**1.0000**	0.8094
Abalone	8	4	2	5	0.9485	**1.0000**	**1.0000**
Satimage	36	10	3	33	**0.8809**	0.7083	0.5950
Yeast	7	6	5	6	**1.0000**	**1.0000**	0.7870
Spambase	57	26	4	49	**0.9176**	0.8950	0.8461
Pendigits	16	13	12	15	0.8600	0.1001	**1.0000**

Table 2. Selected count $|A^\star|$ / Nogueira stability Ψ; p is the original dimension. All five selection methods are shown.

Dataset	p	N-Fscore	AFRSFCV	DTAP	SG-const	SG-dyn
Glass	9	7/0.816	7/0.609	7/0.633	7/0.633	7/0.633
Diabetes	8	7/0.290	3/0.810	7/0.707	7/0.707	7/0.707
Segmentation	18	8/0.885	2/1.000	16/0.800	16/0.888	16/0.888
Abalone	8	4/0.950	2/1.000	5/1.000	5/1.000	5/0.870
Satimage	36	10/0.871	3/0.685	33/0.563	33/0.563	33/0.563
Yeast	7	6/1.000	5/1.000	6/0.757	6/0.757	6/0.757
Spambase	57	26/0.910	4/0.885	49/0.832	48/0.851	48/0.851
Pendigits	16	13/0.854	12/0.063	15/1.000	15/1.000	15/1.000

Former Table B2. Per-dataset classification accuracy (%) and macro one-vs-rest AUC on the held-out test set for RBF-SVM and 5-NN. The best among the three selection methods (N-Fscore, AFRSFCV, DTAP) is in bold.

Dataset	Method	Accuracy (%)		AUC	
		RBF-SVM	5-NN	RBF-SVM	5-NN
Glass	Full	76.74	72.09	0.9250	0.9097
	N-Fscore	69.77	**74.42**	0.9023	0.8933
	AFRSFCV	**72.09**	69.77	0.9078	**0.9371**
	DTAP	**72.09**	69.77	**0.9144**	**0.9371**
Diabetes	Full	74.68	71.43	0.8089	0.7356
	N-Fscore	71.43	70.78	0.7956	0.7154
	AFRSFCV	65.58	67.53	0.6617	0.7145
	DTAP	**74.68**	**73.38**	**0.8098**	**0.7766**
Segmentation	Full	93.29	94.37	0.9937	0.9880
	N-Fscore	80.52	87.01	0.9712	0.9801
	AFRSFCV	78.79	86.58	0.9668	0.9646
	DTAP	**93.29**	**94.59**	**0.9940**	**0.9897**
Abalone	Full	61.60	59.81	0.8102	0.7742
	N-Fscore	61.36	**61.00**	0.8065	0.7605
	AFRSFCV	62.56	58.73	0.8113	**0.7752**
	DTAP	**62.68**	60.29	**0.8123**	0.7695
Satimage	Full	89.12	91.14	0.9796	0.9792
	N-Fscore	87.10	87.80	0.9755	0.9705
	AFRSFCV	80.42	80.96	0.9611	0.9306
	DTAP	**89.12**	**91.22**	**0.9796**	**0.9766**
Yeast	Full	58.25	51.52	0.8480	0.7667
	N-Fscore	**58.92**	**51.52**	0.8057	**0.7763**
	AFRSFCV	57.91	**51.52**	0.8218	0.7681
	DTAP	58.59	**51.52**	**0.8380**	0.7585
Spambase	Full	92.07	90.34	0.9663	0.9443
	N-Fscore	88.38	85.78	0.9391	0.9214
	AFRSFCV	87.62	88.49	0.9155	0.9206
	DTAP	**91.97**	**89.25**	**0.9615**	**0.9437**
Pendigits	Full	99.09	98.95	0.9995	0.9982
	N-Fscore	**98.95**	98.64	**0.9996**	0.9982
	AFRSFCV	97.77	97.73	**0.9996**	0.9965
	DTAP	**98.95**	**99.05**	0.9995	**0.9987**

Table 3. Per-dataset accuracy (%) / macro one-vs-rest AUC on the single holdout. Equal voted masks reuse the same predictions; the SG exceptions are stated below.

Dataset	Classifier	Full	N-Fscore	AFRSFCV	DTAP
Glass	SVM	76.74/0.9222	69.77/0.9073	72.09/0.9174	72.09/0.9174
	5-NN	72.09/0.9097	74.42/0.8933	69.77/0.9371	69.77/0.9371
Diabetes	SVM	74.68/0.8089	71.43/0.7957	65.58/0.6617	74.68/0.8098
	5-NN	71.43/0.7356	70.78/0.7154	67.53/0.7145	73.38/0.7766
Segmentation	SVM	93.29/0.9936	80.52/0.9714	78.79/0.9668	93.29/0.9940
	5-NN	94.37/0.9880	87.01/0.9801	86.58/0.9646	94.59/0.9897
Abalone	SVM	61.60/0.8102	61.36/0.8065	62.56/0.8114	62.68/0.8123
	5-NN	59.81/0.7742	61.00/0.7605	58.73/0.7752	60.29/0.7695
Satimage	SVM	89.12/0.9796	87.10/0.9756	80.42/0.9611	89.12/0.9797
	5-NN	91.14/0.9792	87.80/0.9705	80.96/0.9306	91.22/0.9766
Yeast	SVM	58.25/0.8561	58.92/0.8241	57.91/0.8207	58.59/0.8426
	5-NN	51.52/0.7667	51.52/0.7763	51.52/0.7681	51.52/0.7585
Spambase	SVM	92.07/0.9662	88.38/0.9392	87.62/0.9155	91.97/0.9616
	5-NN	90.34/0.9443	85.78/0.9214	88.49/0.9206	89.25/0.9437
Pendigits	SVM	99.09/0.9995	98.95/0.9996	97.77/0.9996	98.95/0.9995
	5-NN	98.95/0.9982	98.64/0.9982	97.73/0.9965	99.05/0.9987

Both SG variants equal DTAP exactly on the other seven datasets. On Spambase, both give 91.86/0.9628 (SVM) and 89.14/0.9441 (5-NN).

~~Table 2 shows that DTAP retains a moderate number of features. It retains 16/18 on Segmentation, 33/36 on Satimage, and 49/57 on Spambase, whereas AFRSFCV's softmax retains very few features (2, 3, and 4) and N-Fscore lies in between. High stability alone does not imply quality. AFRSFCV is perfectly stable on several sets, yet its accuracy is low because its selections are overly sparse, as shown next.~~

Former Table 4. Classification accuracy (%) and macro one-vs-rest AUC on the held-out test set, averaged over the eight datasets. Per-dataset results are given in Appendix B (Table 3). The best among the three selection methods (N-Fscore, AFRSFCV, DTAP) is in bold.

Method	Accuracy (%)		AUC	
	RBF-SVM	5-NN	RBF-SVM	5-NN
Full	80.61	78.71	0.9164	0.8870
N-Fscore	77.05	77.12	0.8994	0.8770
AFRSFCV	75.34	75.16	0.8807	0.8759
DTAP	**80.17**	**78.63**	**0.9136**	**0.8938**

~~Table 4 reports the averages over the eight datasets, and Table 3 in Appendix B gives the per-dataset results. On average DTAP remains within half an accuracy point of the full feature set, is the best of the three selection methods on all four metrics, and even exceeds the full set on the 5-NN AUC. Per dataset, it matches the full set with RBF-SVM on Segmentation, Satimage, and Diabetes, and exceeds it with 5-NN on Segmentation and Pendigits. Against the base models the advantage is largest on many-class or high-dimensional sets. On Segmentation DTAP exceeds N-Fscore by 12.77 accuracy points and AFRSFCV by 14.50, and on Satimage by 2.02 and 8.70. AFRSFCV's gaps stem from its overly sparse selections, which discard features needed~~

Table 4. Unweighted mean classification performance over the eight classification datasets. All five selection methods and the full-feature reference are included.

Method	SVM accuracy	5-NN accuracy	SVM AUC	5-NN AUC
Full	80.61	78.71	0.9170	0.8870
N-Fscore	77.05	77.12	0.9024	0.8770
AFRSFCV	75.34	75.16	0.8818	0.8759
DTAP	80.17	78.63	0.9146	0.8938
SG-const	80.16	78.62	0.9148	0.8939
SG-dyn	80.16	78.62	0.9148	0.8939

~~for multi-class discrimination, whereas the convex constraints $F_1 \leq b_1, F_2 \leq b_2$ keep DTAP's weights distributed across enough features to preserve the classification structure. The AUC results show the same trend. The largest gap is the SVM-AUC on Diabetes, where AFRSFCV reaches only 0.6617 against DTAP's 0.8098, a direct result of retaining only three features.~~

Former Table 5. Two-sided Wilcoxon signed-rank p-values (DTAP versus each method, over 8 datasets) and Friedman p-values for all five methods.

Metric	Wilcoxon (DTAP vs)				Friedman
	N-Fscore	AFRSFCV	SG-const	SG-dyn	
Accuracy, RBF-SVM	0.031	0.016	1.000	1.000	0.001
Accuracy, 5-NN	0.375	0.031	1.000	1.000	0.006
AUC, RBF-SVM	0.016	0.016	0.031	0.938	0.005
AUC, 5-NN	0.078	0.156	1.000	1.000	0.043

Table 5. DTAP versus each baseline: raw / Holm-adjusted signed-rank p-values. Each metric has four comparisons over eight datasets. The last column gives the tie-corrected Friedman asymptotic p-value for all five methods.

Metric	N-Fscore	AFRSFCV	SG-const	SG-dyn	Friedman
Accuracy, SVM	0.0312/0.0938	0.0156/0.0625	1.0000/1.0000	1.0000/1.0000	0.0005
Accuracy, 5-NN	0.3750/1.0000	0.0312/0.1250	1.0000/1.0000	1.0000/1.0000	0.0057
AUC, SVM	0.0156/0.0625	0.0312/0.0938	1.0000/1.0000	1.0000/1.0000	0.0055
AUC, 5-NN	0.0781/0.3125	0.1562/0.4688	1.0000/1.0000	1.0000/1.0000	0.0435

4.4. Constraint activity and a tuned subgradient comparison

The percentile calibration defines experimental thresholds; it does not guarantee that their sublevel sets intersect. The sampling pool contains 3 000 Dirichlet points and all simplex vertices. The heuristic chooses percentile levels from the sample size, the feature dimension, the number of classes and the class-imbalance ratio; the exact levels and thresholds are recorded in the supplementary metadata. In an application with prescribed thresholds, these values are inputs, and consistency remains an assumption. For the full-data instances below, saved numerical witnesses satisfy all three inequalities strictly under an independent evaluation; the smallest slack is 4.33×10^{-4}. This

Table 6. Optimization evaluations from the uniform simplex start. AC denotes the all-convex comparator; sampled and tuned refer to SG-dyn. † means the tolerance was not reached within the budget. The last column gives the DTAP third-constraint violation count divided by its number of evaluated points.

Dataset	DTAP	AC	SG-const	Sampled	Tuned	F_3 activity
Glass	89	89	520	1000†	100	0/89
Diabetes	14	14	350	1000†	51	0/14
Segmentation	3	3	124	170	3	0/3
Abalone	48	48	638	1000†	3	1/48
Satimage	2	2	60	102	2	0/2
Yeast	4	4	324	305	3	0/4
Spambase	5	5	93	295	56	0/5
Pendigits	2	2	49	10	31	0/2
Movement Libras	637	637	62	1000†	144	0/637

numerical check is distinct from an exact feasibility certificate. All Fisher coordinates are positive on these instances, with minimum 5.99×10^{-5}.

We measure the frequency of $F_3(w^k) - b_3 > 10^{-6}$ over all evaluated points, including the terminating point. This measures violation along the trajectory, rather than whether the constraint is binding at its limit. Table 6 shows that the third track is inactive on seven of the eight standard-start runs for the classification datasets and active at only one of the 48 evaluated points on Abalone. Moreover, the quadratic-over-linear function F_3 is convex when $Q_R \succeq 0$ and $\rho^\top w > 0$. These feature-selection instances therefore test a convex subclass and do not demonstrate a need for the nonconvex extension. We include the valid all-convex comparator, which assigns all three constraints to the convex track. Movement Libras is an additional solver robustness case, separate from the eight-dataset classification study.

Protocol and tuning cost. These are optimization-only tests on full-data feasibility instances. They use $J \le 10^{-6}$, at most 1 000 evaluated points, and identical starting points across methods. SG-const uses $\nu_k = 10^{-3}$. For SG-dyn, $\widehat{L}$ is the largest gradient norm at 500 sampled points. The subgradient implementation follows the most-violated-constraint strategy of [19]. For $D_k = \arg\max_i v_i(x^k)$ it uses

$$x^{k+1} = P_K(x^k - \nu_k \bar{h}_k), \qquad \bar{h}_k = \frac{1}{|D_k|} \sum_{i \in D_k} \frac{\nabla F_i(x^k)}{\|\nabla F_i(x^k)\|},$$

where $\nu_k = 10^{-3}$ for SG-const and $\nu_k = J(x^k)/L$ for SG-dyn. The averaging specifies how ties are handled. The tuned variant selects $L = c\widehat{L}$ from the manually specified grid $c \in \{10^{-3}, 10^{-2.5}, \ldots, 10^{1}\}$ using three random starting points (seeds 1101–1103). Selection first maximizes the number of successful runs, then minimizes total evaluations, then total best residual. Evaluation uses the standard start and three different random starts (seeds 2101–2103). The feature starts are Dirichlet samples. Only starting points are held out: tuning and evaluation use the same full-data instance, and no classification score is used. Neither $\widehat{L}$ nor the tuned value is a certified global Hölder modulus.

Whenever $F_3(w^k) \le b_3$ along a run, the pseudoconvex track is empty, $\mu_k = 1$ and DTAP coincides step by step with the all-convex comparator; on the eight instances

Table 7. Nonconvex pseudoconvex tests, grouped by κ. Each entry is successes out of 12 runs, followed by the median number of evaluated points in parentheses, including capped runs. Each group contains all three dimensions and four starts. Tuned SG-const uses the step selected as described in the text.

κ	DTAP	Balanced	SG-const	Tuned SG-const	Sampled SG-dyn	Tuned SG-dyn
0.3	0 (1000)	0 (1000)	0 (1000)	12 (15)	8 (725.5)	12 (142.5)
1	12 (70)	12 (395.5)	0 (1000)	12 (15)	12 (301.5)	12 (44)
3	12 (4)	12 (4)	0 (1000)	12 (16)	12 (167)	12 (11)

with zero F_3 activity, the identical DTAP and AC columns reflect this identity. On Abalone, F_3 is violated only at the uniform start (1/48), where $J_0^{\mathrm{ps}} \approx 1.56 \times 10^{-3}$ is negligible against $J_0^{\mathrm{cvx}} \approx 366.3$, so $1 - \mu_0 \approx 4.27 \times 10^{-6}$; the values of J along the two trajectories differ by at most 3.18×10^{-5} relative to AC, and both runs stop after 48 evaluated points. The columns are therefore not a comparison of the two assignments, whose success counts differ only from the random starts (26/36 versus 36/36).

Tuning raises SG-dyn's standard-start success count from 5/9 to 9/9. It can outperform DTAP, for example on Abalone, and SG-const is faster on Movement Libras. Across all four starts on all nine feature instances, the success counts are 26/36 for DTAP, 36/36 for the all-convex comparator, 33/36 for SG-const, 15/36 for sampled SG-dyn, and 28/36 for tuned SG-dyn. A balanced DTAP variant also obtains 26/36; it assigns equal weights when both tracks are nonzero and full weight to the sole nonzero track. Thus these tests do not establish a uniform advantage of residual-share weighting or robustness to initialization.

The nine feature-instance searches required 179 744 optimization evaluations and 10.80 seconds of tuning solver time in total. These costs are additional to Table 6; DTAP was not tuned. Initial modulus sampling, preprocessing and search overhead are excluded from this solver-time total. The supplementary records give the cost of each search separately.

A genuinely nonconvex pseudoconvex constraint. To exercise the third track directly, consider $K = [-2, 2]^d$ and

$$\begin{aligned} &F_1(x) = \|x + e_1\|^2 \le 1, \qquad F_2(x) = \|x - e_2\|^2 \le \tfrac{9}{4}, \\ &F_3(x) = \kappa \log(1 + \|x - e_1\|^2) \le \kappa \log 3. \end{aligned} \tag{47}$$

Here $d \in \{2, 10, 50\}$ and $\kappa \in \{0.3, 1, 3\}$ give nine configurations. The third function is pseudoconvex: it is an increasing differentiable transform, with positive derivative, of a convex quadratic. It is not convex, since its radial Hessian eigenvalue is $2\kappa(1-r^2)/(1+r^2)^2 < 0$ for $r = \|x - e_1\| > 1$. The exact witness $x = -e_1/10$ has positive slacks 19/100, 31/25, and $\kappa \log(300/221)$. The standard start is $1.5\mathbf{1}$; the three evaluation starts and three tuning starts are drawn uniformly from $[0.8, 1.8]^d$ with the seeds above (NumPy `default_rng`). For modulus sampling we use 500 uniform points in K (seed 42). SG-const uses the same step $\nu_k = 10^{-3}$ as on the feature instances. Tuned SG-const uses a constant step ν selected from the grid $\{10^{-3}, 10^{-2.5}, \ldots, 10^{1}\}$ with the tuning starts and selection rule of tuned SG-dyn; it selects $\nu = 10^{-0.5}$ in all nine configurations. The balanced variant is the one described above.

DTAP encounters a third-constraint violation in every one of these 36 runs. Its pooled activity is 12 835/12 877 evaluated points. Nevertheless, it reaches the tolerance in only

Table 8. DTAP on (47) with one parameter changed at a time, $\gamma_k = a(k+0.5)^{-s}$. The baseline $\rho_{\mathrm{cvx}} = 1.98$, $\delta = 10^{-3}$, $a = 1$, $s = 0.55$ is the DTAP column of Table 7. Each entry is successes out of 12 runs (three dimensions, four starts), followed by the median number of evaluated points in parentheses, including capped runs.

Setting	$\kappa = 0.3$	$\kappa = 1$
Baseline	0 (1000)	12 (70)
$\rho_{\mathrm{cvx}} = 1$	4 (1000)	12 (223.5)
$\delta = 10^{-1}$	0 (1000)	12 (70)
$a = 10$	12 (77)	12 (6)
$a = 100$	12 (6)	12 (48)
$s = 0.75$	0 (1000)	12 (374)
$s = 1$	0 (1000)	0 (1000)

24/36 runs, compared with 32/36 for sampled SG-dyn, 36/36 for tuned SG-dyn and 36/36 for tuned SG-const (Table 7). Tuned SG-const reaches the tolerance in every run with a median of 15–16 evaluated points, fewer than any other method at $\kappa = 0.3$ and $\kappa = 1$; at $\kappa = 3$, DTAP and the balanced variant need a median of 4. These counts exclude the 147 331 evaluations of its tuning search. With $\nu_k = 10^{-3}$, SG-const moves at most $999 \times 10^{-3} = 0.999$ from its start within the budget, whereas every start lies at distance at least 1.11 from Ω; its zero counts are therefore fixed by the step size and the budget. For $\kappa \le 1$ the convex track is satisfied from the second evaluated point on, so DTAP then follows the single-track iteration of Corollary 3.14; both tracks are violated simultaneously at only 32 of these 12 877 evaluated points. All DTAP failures occur at $\kappa = 0.3$. Changing κ leaves the feasible set unchanged, but v_3 and ∇F_3 both scale with κ, so when $\|g_k^{\mathrm{ps}}\| \le 1$ the pseudoconvex step $(1 - \mu_k)\gamma_k g_k^{\mathrm{ps}}$ scales with $\gamma_k \kappa^2$. Table 8 changes one DTAP parameter at a time for $\kappa \in \{0.3, 1\}$. With $\gamma_k = a(k + 0.5)^{-0.55}$, $a = 10$ and $a = 100$ give 12/12 successes at $\kappa = 0.3$; $\rho_{\mathrm{cvx}} = 1$ gives 4/12, all with $d = 2$; $\delta = 10^{-1}$ reproduces the baseline iterates exactly (Remark 4); the faster-decaying steps $s = 0.75$ and $s = 1$ give no success at $\kappa = 0.3$, and $s = 1$ also fails in all 12 runs at $\kappa = 1$. The failures at $\kappa = 0.3$ are therefore consistent with a diminishing step that is too small for this constraint scale, rather than with a change in feasibility. These tests vary one parameter at a time on one problem family and do not identify a setting that is best in general. The synthetic tuning searches consumed 188 295 optimization evaluations and 10.98 seconds of solver time for SG-dyn, and 147 331 evaluations and 8.27 seconds for SG-const, with the same exclusions as above, in the environment stated below. The experiments support neither universal speed superiority nor a finite-budget robustness claim.

Reproducibility. The feature-selection results were saved with Python 3.11.7 and NumPy 1.26.4 on Windows, with one BLAS/OpenMP thread requested. The feature oracles use algebraically compressed quadratic forms; checked differences from the original values and gradients are at most 1.82×10^{-12} and 1.10×10^{-11}, respectively. The solver times reported for the feature instances refer to this compressed implementation. The accompanying package contains only the uncompressed oracles; it reproduces the evaluation counts, while the reported solver times are those of the original compressed implementation. Tables 7 and 8 and the synthetic tuning costs were produced with Python 3.13.4 and NumPy 2.1.3 on Windows with the default number of BLAS threads; their counts are deterministic, and their solver times are single measurements

on that machine. In this environment the package also reproduces exactly every numerical entry of Tables 1–5 (Tables 4 and 5 from unrounded scores) and of Table 6, and the counts quoted in Sections 4.3 and 4.4. The supplementary scripts, raw trajectories, tuning records, witnesses, source hashes and validation instructions accompany the results.

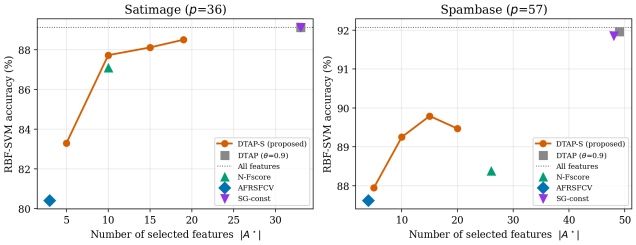
~~Table 5 gives two observations. First, DTAP and the two variants SG-const and SG-dyn are statistically indistinguishable on three of the four metrics: the Wilcoxon p-values equal 1.00 for accuracy on both classifiers and for 5-NN AUC. Their voted selections are in fact identical on seven of the eight datasets, differing only on Spambase (49 features for DTAP against 48 for the SG variants). The tests separate them only on RBF-SVM AUC, where the macro-AUC differs by small margins even when the selection is identical, giving $p = 0.031$ against SG-const and $p = 0.938$ against SG-dyn. This near-equivalence justifies using DTAP as their representative. Second, DTAP outperforms both base models on the SVM metrics. Against N-Fscore, $p = 0.031$ for accuracy and $p = 0.016$ for AUC; against AFRSFCV, $p = 0.016$ for both accuracy and AUC. After Holm–Bonferroni correction over the four comparisons these SVM advantages become borderline (adjusted $p \approx 0.06$), which reflects the small sample of eight datasets, but the Friedman test still rejects equality of the five methods on SVM accuracy ($p = 0.001$), SVM-AUC ($p = 0.005$), 5-NN accuracy ($p = 0.006$), and 5-NN AUC ($p = 0.043$). Overall, DTAP's advantage over the base models is concentrated on the SVM classifier and is weaker for 5-NN.~~

~~[Former Section 6.3] Explicit feature budgets with DTAP-S~~

~~Section 5.5 observed that the cumulative rule lets DTAP (and the SG variants, which share the rule) keep most features on high-dimensional data. We now test the sparse variant DTAP-S on the two highest-dimensional benchmarks, Satimage (p=36) and Spambase (p=57), under the identical 80/20 split and majority-vote-over-ten-folds protocol, sweeping the budget s. Table 6 and Fig. 1 report the outcome.~~

~~**Former Table 6.** Sparse selection with DTAP-S on the two highest-dimensional benchmarks, reporting selected size $|A^\star|$, held-out accuracy (%), and Nogueira stability Φ. The budget s in DTAP-S$_s$ is an explicit cap on $|A^\star|$. Best accuracy at a matched small budget is in bold.~~

Dataset	Method	$\lvert A^\star\rvert$	RBF-SVM	5-NN	Φ
Satimage (p=36)	Full	36	89.12	91.14	–
	N-Fscore	10	87.10	87.80	0.881
	AFRSFCV	3	80.42	80.96	0.708
	DTAP	33	89.12	91.22	0.595
	DTAP-S$_{15}$	15	88.11	89.98	0.852
	DTAP-S$_{10}$	10	**87.72**	**89.43**	0.852
Spambase (p=57)	Full	57	92.07	90.34	–
	N-Fscore	26	88.38	85.78	0.918
	AFRSFCV	4	87.62	88.49	0.895
	DTAP	49	91.97	89.25	0.846
	DTAP-S$_{10}$	10	**89.25**	**88.71**	1.000
	DTAP-S$_5$	5	87.95	88.71	1.000



Former Figure 1. RBF-SVM accuracy versus selected size $|A^\star|$ on Satimage and Spambase. At similar selected sizes the DTAP-S curve (orange) lies above the baseline points, and on Spambase DTAP-S_{10} outperforms N-Fscore with fewer features and higher accuracy.

~~On Satimage and Spambase, where a small set of useful features exists, DTAP-S makes DTAP select a genuinely small set and outperforms the baselines at similar budgets. On Spambase, DTAP-S_{10} uses only 10 features yet reaches 89.25/88.71 (SVM/5-NN) with perfect stability $\Phi = 1.000$, while N-Fscore needs 26 features for the lower 88.38/85.78 at $\Phi = 0.918$. DTAP-S_5 with 5 features likewise outperforms AFRSFCV with 4 features on all three measures. On Satimage at a matched 10 features, DTAP-S_{10} (87.72/89.43) again exceeds N-Fscore (87.10/87.80). The budget also cuts DTAP's own selection from 33/49 features to 10 while *raising* stability from 0.595/0.846 to 0.852/1.000, since a fixed budget forces a consistent set across folds (Fig. 1).~~

~~[Former Section 6.4] Runtime and convergence~~

~~We now compare the computational cost and convergence behaviour of the five methods. Table 7 reports the wall-clock time, iteration count, and per-iteration cost of each method, averaged first over the cross-validation folds and then over the eight datasets; the per-dataset values are given in Appendix B (Table B3).~~

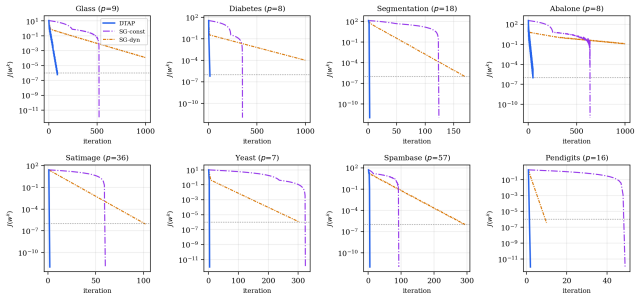
Former Table 7. Wall-clock time, iteration count, and per-iteration cost, averaged over the eight datasets (fold averages per dataset). The time averages are dominated by the largest datasets; per-dataset values are in Appendix B (Table B3). Best per column in bold.

Method	Time (s)	Iterations	Cost (ms/iter)
N-Fscore	0.057	999.0	**0.057**
AFRSFCV	1.620	1000.0	1.620
SG-const	0.180	300.7	1.758
SG-dyn	0.883	484.8	25.569
DTAP	**0.007**	**25.1**	0.935

~~Table 7 shows that DTAP has the lowest runtime, and the per-dataset values in Table B3 confirm this on every one of the eight datasets. The advantage does not come from the per-iteration cost. As noted in Remark 6, DTAP, AFRSFCV, and SG-const share the same $O(nrp + p^2)$ per-iteration order, since all three compute the constraint gradients on the same input matrices. On Satimage, for instance, they measure 2.587, 4.140, and 4.492 ms per iteration. The advantage comes from the iteration count. Thanks to the stopping test $J(w^k) \le \varepsilon$, DTAP stops after very few iterations, whereas AFRSFCV uses the full 1 000-iteration budget and SG-dyn often reaches it too. The gap is largest on the large sets. On Pendigits DTAP needs 0.004 s versus 4.893 s for AFRSFCV and~~

1.761 s for SG-dyn, and on Satimage the three are 0.005, 4.140, and 2.032 s. N-Fscore runs 999 low-cost Euler steps that do not recompute the residual on large matrices, but it is still slower than DTAP on all eight sets.

A separate convergence experiment on the full data of each set gives a consistent picture (Fig. 2). DTAP and SG-const drive J below ε on all eight datasets, whereas SG-dyn reaches feasibility on only five. The reason is that the estimated Hölder modulus is often very large, about 6.5×10^3 on Segmentation for instance, which makes the dynamic step tiny and convergence slow. This agrees with Theorem 3.12 on the strong convergence of DTAP and shows the benefit of the Polyak step on the convex track. DTAP avoids estimating the Hölder modulus and still reaches feasibility in few iterations.



Former Figure 2. Violation $J(w^k)$ (log scale) versus iteration for DTAP, SG-const, and SG-dyn on the eight datasets. DTAP (solid) reaches feasibility in very few iterations on every dataset, while SG-dyn (dashed) stalls on several because the estimated Hölder modulus makes its step size too small.

[Former Section 6.5] Further analyses of robustness, fidelity, and ablation

Five additional experiments examine the reasons behind the results above; we summarise them briefly. *(i) Constant-free robustness.* Every subgradient baseline must fix a step constant, namely v for SG-const and the Hölder modulus L for SG-dyn, and it reaches feasibility only inside a narrow band of that constant. Averaged over five datasets and nine swept values, SG-const succeeds for 6.2/9 values and SG-dyn for 3.8/9, whereas DTAP carries no such constant and reaches feasibility from the single standard start on all eight datasets. *(ii) Deterministic convergence.* From the standard start DTAP reaches $J \le \varepsilon$ on all eight sets, SG-const does so as well but with 5–81× more iterations, and SG-dyn succeeds on only five because it stalls when the estimated modulus shrinks its step. The DTAP trajectory also has no run-to-run variance. *(iii) Performance profile.* On iteration count DTAP is fastest on all eight datasets ($\rho(1) = 1.0$), which confirms that the speed-up of Section 6.4 comes from the iteration count, not the per-iteration cost. *(iv) Selection fidelity on synthetic ground truth.* On controlled data (n=600, p=30, four classes, eight informative features), DTAP recovers all informative features at every redundancy level, rejects the most noise, and reaches the highest precision of the three selection methods, whereas the N-Fscore PNN under-selects and discards 12–44% of the informative features. *(v) Dual-track ablation.* Collapsing DTAP to a single track supports the design. A single diminishing-step subgradient on all constraints fails to reach feasibility within budget on four of five datasets, and a Polyak-only variant has no guarantee once the pseudoconvex constraint binds; only the full dual-track scheme is both guaranteed and fast. A Nemenyi critical-difference test on SVM accuracy ranks

DTAP first (mean rank 2.17), significantly ahead of AFRSFCV.

[Former Section 6.6] Summary of findings

The numerical study supports four conclusions. (i) *Quality.* DTAP matches or exceeds the full-feature accuracy and AUC on most datasets and outperforms both base models on the SVM metrics (Tables 4 and 5). (ii) *Selection size.* The two convex constraints retain enough features to preserve the multi-class structure, in contrast to the overly sparse softmax selections of AFRSFCV (Table 2). (iii) *Speed.* DTAP is the fastest method on all eight datasets, by up to three orders of magnitude on the large sets, and the gain comes from the iteration count rather than the per-iteration cost (Table 7). (iv) *Robust convergence.* DTAP reaches feasibility on all eight datasets, whereas SG-dyn does so on only five because its Hölder-modulus estimate makes the step size too small (Fig. 2). Finally, DTAP and the Hu variants give nearly identical selections (Wilcoxon $p = 1$ on three of the four metrics, with only a small difference on RBF-SVM AUC), so the split-feasibility formulation delivers the quality, while the DTAP design delivers the speed.

5. Conclusion

We studied normal-box nonlinear split feasibility with convex and pseudoconvex component functions. DTAP combines a residual-scaled convex step with a safeguarded pseudoconvex step and projects onto the base set. Its whole-sequence convergence follows from quasi-Fejér descent and the existence of one feasible cluster point. No prescribed regularity modulus is needed to execute the method, but continuity, consistency, and local bounds remain essential to the analysis. The proportional weight is an admissible residual-responsive choice, not an optimality result. The feature-selection criterion with a positive-semidefinite quadratic numerator and positive affine denominator is convex, so that application illustrates a convex special case. The activity measurements and controlled nonconvex tests distinguish these settings. Tuned subgradient methods can match or outperform DTAP, and initialization and scaling affect performance within a finite budget. The best-iterate bound in Remark 5 concerns only the convex-track residual. Rates for the full residual and performance guarantees under additional geometric assumptions remain open.

We studied a nonlinear split feasibility problem with a normal-box output set and mixed convex–pseudoconvex component maps. A Tchebyshev scalarization reduced feasibility to a closed-form max-type violation function, and we proposed DTAP, a dual-track algorithm that takes a residual-adaptive step on the convex constraints and a safeguarded diminishing step on the pseudoconvex ones, fuses the two by a soft weight, and projects only onto the base set. On the convex track the step is the constant-free rule of López et al. [25] extended to the nonlinear setting, licensed now by convexity of the individual components rather than by firm nonexpansiveness of a linear composition; the pseudoconvex track absorbs the remaining constraints on the sign that pseudoconvexity still supplies. Its step sizes need no operator-norm, Lipschitz, or Hölder constant, and we proved strong convergence to a feasible point (Theorem 3.12). Two predecessors are closest. Against the nonlinear multiple-sets solver of Li et al. [23], DTAP uses the same search direction but replaces the global, unverifiable hypothesis that the aggregate proximity function be convex (a hypothesis that a single pseudoconvex component invalidates, leaving the backtracking search possibly undefined) by a per-constraint classification and a closed-form step, and it drops the boundedness and global Lipschitz

requirements of that analysis. Against the subgradient method of Yen et al. [37], with which it shares the max-type scalarization and the active-set Clarke direction, DTAP differs in the admissible class (convex-plus-pseudoconvex, which neither contains nor is contained in their quasilinear class) and in driving the convex part by a non-diminishing, residual-scaled step rather than a prescribed diminishing sequence (Table 1).

On the application side, two previously separate feature-selection models reduce to a single normal-box NSFP whose constraints are drawn from their objectives, and the sparse variant DTAP-S adds an explicit feature budget. On eight UCI benchmarks DTAP preserved or improved accuracy and AUC over the base models, with the clearest gains on the SVM metrics, and reached feasibility substantially faster than the same-class subgradient methods, while DTAP-S gave compact, stable selections that outperform the baselines at matched size. This answers the three questions posed in the introduction.

Several limitations should be stated plainly. The theory delivers a feasibility guarantee rather than optimality, with a worst-case best-violation rate of $O(N^{-1/3})$ (Remark 5), and sharper rates need extra structure such as an error bound. Performance depends on the threshold calibration of Section 6.1.3, since the percentiles are tuned per dataset and a poor choice can empty Ω; the calibration is applied identically to the whole split-feasibility group, so it cannot favour DTAP over its same-class competitors. The iteration count can grow with the feature dimension, and the $p \gg n$ regime of text or genomic data is untested. Finally, we report two classifiers and two metrics on eight datasets, a modest sample, so we treat marginal tests such as the 5-NN AUC as inconclusive.

Because criteria enter the model as constraints, further objectives such as sparsity surrogates or fairness penalties can be appended as convex or pseudoconvex constraints without changing the solver. An inertial acceleration would control the asymptotic regularity of Proposition 3.11, and the Hilbert-space analysis points to infinite-dimensional extensions and to other applications of nonlinear split feasibility, such as equilibrium models [37] and constrained learning [31].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI use

The authors report that generative AI was not used in the research or the preparation of this manuscriptOpenAI Codex was used during revision for language editing.

Data availability

The ~~eight datasets used in this study are publicly~~ datasets are available from the UCI Machine Learning Repository [21]. ~~An implementation of and the baseline solvers is available from the authors on reasonable request~~The accompanying reproducibility package contains the scripts, recorded numerical outputs, and instructions for regenerating the reported tables.

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~~[Former Section A] Baseline solvers~~

[This paragraph of the submitted version (former Section A) was moved; see p. 32]

[This paragraph of the submitted version (former Section A) was moved; see p. 32]

[This paragraph of the submitted version (former Section A) was moved; see p. 33]

~~[Former Section B] Experimental details~~

~~This appendix collects the threshold calibration, the implementation details, and the per-dataset classification and runtime results deferred from Section 4.~~

~~**Output-threshold calibration** For each dataset we evaluate F_1, F_2, F_3 at the 3 000 sampled points $w^{(s)} \sim \mathrm{Dir}(\mathbf{1}_p)$ and set b_i to the p_i-th percentile of the resulting values, so about $p_i\%$ of simplex points satisfy $F_i \le b_i$ and the percentile controls the size of the feasible set. The percentiles p_i are tuned per dataset (by sample size, dimension, class count, and balance). The resulting b_i, averaged over folds, appear in Table B1.~~

Former Table B1. Output thresholds b_1, b_2, b_3 (fold averages) per dataset.

Dataset	b_1	b_2	b_3
Glass	142	18.3	0.0638
Diabetes	549	98.9	0.0443
Segmentation	1 450	466	0.00956
Abalone	2 990	1 080	0.105
Satimage	4 570	3 200	0.0356
Yeast	887	579	0.0524
Spambase	3 250	1 010	0.0800
Pendigits	7 910	7 080	0.0141

~~**Implementation and reproducibility** All five methods share one per-fold pipeline and consume identical precomputed inputs $(R_k(D), P, Y, Q, \rho)$. Min–max scaling is fit~~

on each fold's training portion only, multi-information uses 10-bin equal-width histograms [4], Q is shifted by $\delta_Q = -\min\{0, \lambda_{\min}(S)\}$ to keep $Q \succeq 0$, the Fisher denominator uses $\varepsilon_\rho = 10^{-8}$, and the simplex projection is the sort-based $O(p \log p)$ algorithm [11]. Random seeds for the CV splits and the Dirichlet sampling are fixed, so the study is reproducible. Reported runtimes are relative cross-method measurements and are not sensitive to hardware, because the per-iteration cost is dominated by the shared matrix operations.

Per-dataset classification results Table 3 reports the per-dataset accuracy and macro one-vs-rest AUC that underlie the averages of Table 4.

[This paragraph of the submitted version (former Section B) was moved; see p. 36]

Per-dataset runtime Table B3 reports the per-dataset wall-clock time, iteration count, and per-iteration cost that underlie the averages of Table 7.

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Former Table B3. Wall-clock time, iteration count, and per-iteration cost (fold averages) of the five methods.

Dataset	Method	Time (s)	Iterations	Cost (ms/iter)
Glass	N-Fscore	0.059	999.0	**0.059**
	AFRSFCV	0.160	1000.0	0.160
	SG-const	0.101	503.1	0.201
	SG-dyn	0.265	991.4	0.267
	DTAP	**0.012**	**75.7**	0.148
Diabetes	N-Fscore	0.057	999.0	**0.057**
	AFRSFCV	0.166	1000.0	0.166
	SG-const	0.070	324.7	0.216
	SG-dyn	0.245	923.6	0.272
	DTAP	**0.002**	**11.7**	0.169
Segmentation	N-Fscore	0.056	999.0	**0.056**
	AFRSFCV	0.712	1000.0	0.712
	SG-const	0.098	124.5	0.786
	SG-dyn	0.341	159.7	2.133
	DTAP	**0.003**	**4.9**	0.557
Abalone	N-Fscore	0.055	999.0	**0.055**
	AFRSFCV	0.289	1000.0	0.289
	SG-const	0.284	881.8	0.322
	SG-dyn	0.415	1000.0	0.415
	DTAP	**0.021**	**88.0**	0.243
Satimage	N-Fscore	0.058	999.0	**0.058**
	AFRSFCV	4.140	1000.0	4.140
	SG-const	0.274	61.0	4.492
	SG-dyn	2.032	107.0	18.998
	DTAP	**0.005**	**2.0**	2.587
Yeast	N-Fscore	0.055	999.0	**0.055**
	AFRSFCV	0.240	1000.0	0.240
	SG-const	0.099	365.0	0.271
	SG-dyn	0.180	377.0	0.501
	DTAP	**0.002**	**11.2**	0.174
Spambase	N-Fscore	0.063	999.0	**0.063**
	AFRSFCV	2.357	1000.0	2.357
	SG-const	0.265	97.0	2.730
	SG-dyn	1.827	309.8	5.902
	DTAP	**0.008**	**5.0**	1.513
Pendigits	N-Fscore	0.056	999.0	**0.056**
	AFRSFCV	4.893	1000.0	4.893
	SG-const	0.246	48.7	5.043
	SG-dyn	1.761	10.0	176.065
	DTAP	**0.004**	**2.0**	2.086

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